HANOI UNIVERSITY OF SCIENCE AND TECHNOLOGY

# GRADUATION THESIS

## Student's engagement detection in online classes


**PHAN MINH HÒA**
hoa.pm225495@sis.hust.edu.vn


**Program: Data Science and Artificial Intelligence**

**Supervisor:** PhD. Trịnh Văn Chiến

**Department:** Computer Science

**School:** School of Information and Communications Technology

**HANOI, 06/2026**

**HANOI UNIVERSITY OF SCIENCE AND TECHNOLOGY**

# GRADUATION THESIS

## Student's engagement detection in online classes

**PHAN MINH HÒA**
hoa.pm225495@sis.hust.edu.vn

**Program: Data Science and Artificial Intelligence**

**Supervisor:** PhD. Trịnh Văn Chiến ____________________

Signature

**Department:** Computer Science

**School:** School of Information and Communications Technology

**HANOI, 06/2026**

# ACKNOWLEDGMENT

I am grateful to my supervisor, **PhD. Trịnh Văn Chiến**, for the guidance, careful feedback, and patience that shaped this thesis. I thank the lecturers and staff of the School of Information and Communications Technology at Hanoi University of Science and Technology for the foundation they gave me, and the creators of the CMOSE [1] and DAiSEE [2] datasets, whose work made this study possible. Finally, I thank my family and friends for their constant support, and acknowledge my own persistence in seeing this work through.

# ABSTRACT

Online and hybrid education has made the webcam the main link between student and teacher, yet a teacher facing many video tiles cannot tell when a learner loses focus. Automatic engagement recognition addresses this by classifying a short webcam clip on an ordinal scale from disengaged to engaged. The task is hard because the labels are ordinal and severely imbalanced toward the engaged levels, and each student is recorded at a different camera angle. Prior work either encodes each frame with a facial-behaviour toolkit or learns directly from pixels, and reports accuracy on a single dataset. This thesis identifies two limits. The mixed facial features are passed through a single encoder, and accuracy is used as the primary metric, which on imbalanced data can look healthy while the model fails to generalise. This thesis keeps the structure of the facial signal instead of flattening it, because gaze, landmarks, head pose, and action units move on different timescales. The proposed hybrid model splits the OpenFace features into feature groups, encodes each group with its own temporal encoder, and optionally adds a global I3D motion stream. Every model is judged by an ordinal, imbalance-aware measure rather than accuracy. An ablation over all encoder assignments shows the proposed hybrid model outperforms the strongest single-encoder baseline. A test-only, in-the-wild private set was collected and hand-labelled, and a cross-dataset study shows that engagement models transfer poorly and that accuracy conceals this collapse. Trained on the combined datasets, the proposed hybrid model generalises best to the unseen private set.

**Keywords:** student engagement recognition, proposed hybrid model, feature-group temporal fusion, ordinal classification, cross-dataset generalisation.

Student

*(Signature and full name)*

## TABLE OF CONTENTS

**CHAPTER 1. INTRODUCTION** .......... **1**

1.1 Problem Statement .......... 1

1.2 Background and Problems of Research .......... 2

1.3 Research Objectives and Conceptual Framework .......... 3

1.4 Contributions .......... 4

1.5 Organization of Thesis .......... 4

**CHAPTER 2. LITERATURE REVIEW** .......... **6**

2.1 Scope of Research .......... 6

2.2 Related Work .......... 6

2.3 Background Knowledge .......... 7

  2.3.1 Feature Extraction .......... 7

  2.3.2 Temporal Architectures .......... 8

  2.3.3 Loss Functions for Imbalance and Ordinality .......... 11

  2.3.4 Evaluation Metrics .......... 12

  2.3.5 Statistical Measures for Result Analysis .......... 14

2.4 Chapter Summary .......... 15

**CHAPTER 3. METHODOLOGY** .......... **16**

3.1 Overview .......... 16

3.2 Data and Preprocessing .......... 17

  3.2.1 Public datasets .......... 17

  3.2.2 Private dataset .......... 18

  3.2.3 Preprocessing .......... 19

3.3 Baseline Models .......... 19

3.4 Proposed Hybrid Model ........................................................................ 21
3.4.1 Grouping the facial features by meaning...................................... 21
3.4.2 Model architecture ....................................................................... 22
3.5 Chapter Summary.......................................................................................... 24
**CHAPTER 4. EXPERIMENTAL RESULTS .......................................... 25**
4.1 Overview ......................................................................................................... 25
4.2 Experimental Setup.......................................................................................... 25
4.2.1 Loss Functions............................................................................... 26
4.2.2 Training Protocol ........................................................................... 26
4.2.3 Metrics and the 3×3 Evaluation Matrix........................................ 27
4.3 Evaluation Properties of the Proposed Model ........................................... 27
4.4 Encoder Choice per Group ......................................................................... 28
4.5 Effect of I3D Fusion ...................................................................................... 29
4.6 Comparison with the Baseline ................................................................... 30
4.7 Cross-Dataset Generalisation .................................................................... 31
4.7.1 Cross-Dataset Matrix...................................................................... 31
4.7.2 The Private Set ............................................................................... 31
4.7.3 In-Domain versus Transfer ........................................................... 33
4.8 Discussion ..................................................................................................... 35
4.9 Chapter Summary.......................................................................................... 36
**CHAPTER 5. CONCLUSIONS ............................................................... 37**
5.1 Summary ......................................................................................................... 37
5.2 Synthesis ....................................................................................................... 38
5.3 Revisiting the Research Questions ........................................................... 39
5.4 Limitations..................................................................................................... 40
5.5 Practical Recommendations........................................................................ 41

5.6 Suggestion for Future Works ............ 41

5.7 Closing Remarks ............ 42

**REFERENCE ............ 44**

**APPENDIX A. BASELINE MODELS ............ 45**

A.1 Experimental Setup ............ 45

A.2 Evaluation Protocol ............ 45

A.2.1 Selecting the Primary Metric ............ 46

A.2.2 Selecting the Development Dataset ............ 47

A.2.3 Selecting the Development Loss ............ 48

A.3 In-Domain Results ............ 48

A.4 Prediction Agreement ............ 49

A.4.1 Agreement between Configurations ............ 50

A.4.2 OpenFace versus I3D ............ 51

A.5 Cross-Dataset Generalisation ............ 51

A.5.1 The $3\times3$ Matrix ............ 51

A.5.2 Closing the Loop: Metric and Loss Reliability ............ 53

A.5.3 In-Domain versus Transfer ............ 54

A.6 Chapter Summary ............ 54

# LIST OF FIGURES

Figure 2.1 Residual TemporalBlock . . . . . . . . . . . . . . . . . . . . . . 9
Figure 2.2 LSTM cell at time step $t$ . . . . . . . . . . . . . . . . . . . . . 10
Figure 2.3 Transformer encoder layer . . . . . . . . . . . . . . . . . . . . . 10

Figure 3.1 The baseline and the proposed hybrid model. . . . . . . . . . . 16
Figure 3.2 Representative frames from the private set, illustrating the range of camera angles found in unconstrained online-class recordings. 18
Figure 3.3 The five single-encoder baselines. . . . . . . . . . . . . . . . . 20
Figure 3.3 The five single-encoder baselines (continued). . . . . . . . . . 21
Figure 3.4 Proposed hybrid model architecture (I3D stream enabled; the I3D-stream-disabled variant drops the I3D row). . . . . . . . . . 22
Figure 3.5 The three interchangeable per-group encoders. . . . . . . . . 23

Figure 4.1 Ablation of the proposed hybrid model across all 243 per-group encoder configurations, with the I3D stream enabled and disabled, on all six metrics (in-domain CMOSE). The dashed line is the QWK-selected baseline (OpenFace TCN, cross-entropy). 28
Figure 4.2 Cross-domain generalisation, best proposed-model configuration per cell (selected by QWK), on QWK, macro-accuracy, macro-MAE, and accuracy. . . . . . . . . . . . . . 32
Figure 4.3 Cell-by-cell advantage of the proposed model: best proposed-model minus best baseline QWK over the 3×3 train×test matrix (Figure 4.2 minus Figure A.5). . . . . . . . . . . . . . . . . 33
Figure 4.4 Private-set confusion of the combined-trained best baseline versus best proposed-model configuration (row-normalised). . . . . 34
Figure 4.5 In-domain QWK (trained and tested on CMOSE) versus transfer QWK (trained on Combined, tested on the held-out private set); each configuration is one point pairing its own two runs. Spearman $\rho = 0.27$ for the 486 proposed-model configurations and $0.31$ for the 15 baselines. . . . . . . . . . . . . . . . . . . . . . . . 35

Figure A.1 Rank agreement (Kendall $\tau$) between the six metrics over the fifteen baseline configurations, in-domain on CMOSE (MAE metrics sign-flipped so positive $\tau$ means agreement). . . . . . . . 47

Figure A.2 QWK against the balanced secondary metrics (macro-accuracy and macro-MAE) for every baseline across the three losses, in-domain on CMOSE. . . . . . . . . . . . . . . . . . . . 48
Figure A.3 The five baseline models under the three losses on all six metrics, in-domain on CMOSE. . . . . . . . . . . . . . . . . . . . . . . 49
Figure A.4 Pairwise prediction agreement (Cohen's $\kappa$) between all fifteen baseline configurations on the in-domain CMOSE test set, in model-major blocks. . . . . . . . . . . . . . . . . . . . . . . . . . 50
Figure A.5 Cross-domain generalisation, best baseline per cell (selected by QWK), on QWK, macro-accuracy, macro-MAE, and accuracy. . 52
Figure A.6 Architecture QWK ranking, in-domain (left; trained and tested on CMOSE) versus the mean over the five unseen-target cells (right), best over loss. One line per architecture, with the rank (1 = best) at each end. . . . . . . . . . . . . . . . . . . . . . . . . . 53

# LIST OF TABLES

Table 3.1 Dataset statistics and class balance (%). . . . . . . . . . . . . . 17
Table 3.2 Grouping of the 709-dimensional OpenFace descriptor into five feature groups by meaning. . . . . . . . . . . . . . . . . . . . . . . 21

Table 4.1 Per-group marginal effect of encoder choice on mean QWK, in-domain (CMOSE) and pooled over the unseen-target cells. . . . . 29
Table 4.2 Paired effect of the I3D stream on QWK, by evaluation regime. 29
Table 4.3 Top-5 configurations of the proposed hybrid model, in-domain (CMOSE), sorted by QWK. . . . . . . . . . . . . . . . . . 30
Table 4.4 Private set (test-only): best baseline vs best proposed hybrid model by training source. . . . . . . . . . . . . . . . . . . . . . . . . 32

Table A.1 Winning baseline configuration by metric (in-domain CMOSE). 46
Table A.2 Best in-domain result per dataset (CMOSE vs DAiSEE). . . . 47
Table A.3 Best OpenFace encoder versus the I3D MLP (in-domain CMOSE, cross-entropy). . . . . . . . . . . . . . . . . . . . . . . . . 51

## LIST OF ABBREVIATIONS

| Abbreviation | Definition |
|---|---|
| AU | Facial Action Unit |
| BiLSTM | Bidirectional Long Short-Term Memory network |
| CDF | Cumulative Distribution Function |
| CE | Cross-Entropy loss |
| CMOSE | Class-level Multimodal Online Student Engagement dataset |
| DAiSEE | Dataset for Affective States in E-Environments |
| EMD | Earth Mover’s Distance; here the squared-EMD / CDF-MSE ordinal loss |
| GELU | Gaussian Error Linear Unit |
| I3D | Inflated 3D ConvNet (spatiotemporal video feature extractor) |
| LSTM | Long Short-Term Memory network |
| MAE | Mean Absolute Error |
| MLP | Multi-Layer Perceptron |
| QWK | Quadratic-Weighted Kappa |
| ReLU | Rectified Linear Unit |
| SMOTE | Synthetic Minority Over-sampling Technique |
| TCN | Temporal Convolutional Network |

# CHAPTER 1. INTRODUCTION

## 1.1 Problem Statement

Online and hybrid education is now a permanent part of teaching, and lectures, tutorials, and exams are given through video calls in which the webcam is the main link between student and teacher. The webcam stream shows how well a student follows the material, but a teacher facing dozens of simultaneous video tiles cannot notice when one learner loses focus. Automatic engagement recognition addresses this problem: it classifies a short webcam clip of one student into one of four ordinal levels, from E0 (Highly-Disengage) to E3 (Highly-Engage), so a teacher or platform can adapt the pace of a lesson.

This thesis adopts the task form used by prior work and by the datasets it builds on. The input is a short front-webcam clip of one student, about ten seconds long. The output is one of four ordinal levels, E0 (Highly-Disengage), E1 (Disengage), E2 (Engage), and E3 (Highly-Engage), a scale from full withdrawal to active attention.

Although it resembles ordinary video classification, engagement recognition is harder for three reasons that shape every design choice in this thesis.

(i) **Ordinal labels.** The four classes lie on a line, $E0 < E1 < E2 < E3$, so predicting E3 for an E0 student is a serious error while predicting E2 for an E3 student is almost right. Accuracy ignores this order and treats both mistakes as equally wrong, so the size of an error must be weighted by the distance between the true and predicted levels.

(ii) **Severe imbalance.** Real classes are mostly attentive, so the datasets are dominated by the upper levels: about 69% of CMOSE clips are E2 and fewer than 3% are E0 (Section 3.2, Table 3.1). A model can therefore reach high accuracy by always predicting the majority class and never flagging the rare disengaged cases a monitoring system must catch.

(iii) **Unfixed camera angle.** Engagement is read from facial behaviour, but the camera geometry differs from learner to learner: a laptop camera looks up from the desk, a monitor camera sits near eye level, a phone is propped to the side. The OpenFace gaze, head-pose, and landmark features are measured in image geometry, so a change of angle moves the features even when the underlying engagement is identical. Each dataset covers its own narrow range of setups, so a deployed model meets unfamiliar viewpoints immediately, yet robustness to this shift is rarely measured.

These three properties follow from how an engagement signal is used, and together they define what a useful model must do. The unit of decision is rarely a single clip but a summary, such as an engagement curve for one student over a session or a class-level spread the teacher watches live. The value of that signal lies in the rare cases: confirming that an attentive class is attentive tells the teacher little, while flagging the few moments a student disengages is the goal, and this is the case that imbalance hides and accuracy does not reward. A deployment also never sees the data the model was tuned on, since cameras, lighting, participants, and rubric all change. The main concern of this thesis is therefore not overall accuracy but ordinal agreement on imbalanced data, and how well it holds when the data shifts, judged on a self-collected, never-trained-on test set under a cross-dataset protocol.

## 1.2 Background and Problems of Research

Vision-based engagement recognition has taken two routes. The first encodes each frame with a facial-behaviour toolkit, most often OpenFace [3], which estimates gaze direction, head pose, facial landmarks, and action-unit intensities, and feeds the feature sequence to a classical classifier or a recurrent network such as an OpenFace+BiLSTM pipeline [4] or classical models on summary features [5]. The second learns directly from pixels with deep video models, for instance an EfficientNetV2 backbone followed by an LSTM [6], or borrows context-modelling ideas from affect recognition [7]. This thesis builds on two in-the-wild datasets: CMOSE [1], a recent, large, expert-labelled dataset that ships ready OpenFace and I3D features, and DAiSEE [2], a widely used public dataset that gives only raw video, from which the same features are extracted. A systematic review [8] confirms that class imbalance and subjective labels are common, unsolved problems across the field.

Two limits of this work motivate the thesis. The first is that the mixed facial-behaviour features are passed through a single encoder. A single OpenFace frame in CMOSE is a 709-dimensional vector that bundles signals of different physical kinds: a few gaze values, hundreds of eye- and face-landmark coordinates, head-pose parameters, and action-unit strengths. These parts move at different speeds, since the eyes flick fast and noisily, head turns are slow and smooth, and action units fire in short bursts. Joining all 709 numbers and pushing them through one sequence encoder forces a single temporal model onto signals that do not share a timescale, and hides the role of each behavioural channel.

The second limit is that studies rely on accuracy as the primary metric and lack a reliable evaluation of generalisation. Almost every study reports accuracy in-domain, training and testing on the same dataset, yet the only deployment that matters is on data recorded with different cameras and conditions. Because engagement labels are subjective and the class balance differs sharply between datasets, models can be expected to transfer poorly, but without measuring transfer this stays an assumption. On imbalanced data a model that predicts the majority class can reach a decent accuracy while learning nothing that transfers, so the metric most papers report hides the failure it should reveal.

These two limits define the gap. The mixed-feature problem calls for a representation that keeps the structure of the facial features instead of flattening it, and the generalisation problem calls for a protocol that measures transfer directly with metrics that majority prediction cannot fool.

## 1.3 Research Objectives and Conceptual Framework

The goal of this thesis is to improve, and describe accurately, how engagement is recognised on a four-level ordinal scale from facial-behaviour and motion features. This thesis pursues it through one architecture idea and one set of evaluation rules, stated as four research questions.

(i) **RQ1: Decomposition.** Does splitting the OpenFace features into groups, each with its own temporal encoder, improve engagement classification over single-encoder baselines that encode all 709 features together?

(ii) **RQ2: Encoder assignment.** If splitting helps, which group→encoder choices matter? Does any behavioural channel prefer one encoder family (convolutional, recurrent, or attention-based)?

(iii) **RQ3: Motion fusion.** Does adding a global I3D motion stream help, and does any gain hold across many configurations rather than one favourable one?

(iv) **RQ4: Generalisation and losses.** How well do these models generalise across datasets, and how does the loss trade ordinal agreement against minority-class recall and balanced ordinal error?

The idea behind RQ1–RQ3 is to treat a clip as several streams in time rather than one sequence, realised by the proposed hybrid model of Chapter 3 (Figure 3.4). The OpenFace features are split into five feature groups (gaze, eye landmarks, face landmarks, head pose, and action units), and each group is encoded on its own into a fixed-length embedding by an encoder suited to its timing. A global I3D clip embedding can be added as a sixth stream. The embeddings are joined and classified, and every stream also gets its own small head so each channel is useful on its own. This design is the proposed hybrid model, the main object tested against the single-encoder baselines. The rule behind RQ4 is to select and rank every model on quadratic-weighted kappa (QWK), an ordinal-agreement metric, reporting balanced accuracy and balanced ordinal error (macro-MAE) only as secondary checks, and to test every model not only in-domain but on three datasets, one collected for this thesis.

## 1.4 Contributions

This thesis makes three contributions.

(i) **Architecture.** This thesis proposes a hybrid model that splits the 709 OpenFace features into five groups, encodes each group with its own TCN, Transformer, or LSTM, optionally adds a global I3D motion stream, and trains every stream with an auxiliary head (Section 3.4). Trying all 243 encoder assignments, with the I3D stream enabled and disabled, shows the I3D-stream-enabled configurations score higher than the strongest baseline (median QWK 0.553 against 0.537, best 0.605), that the gain belongs to the family rather than one configuration, and that head pose is the only group with a clear encoder preference, for a TCN, which holds under shift (Section 4.3).

(ii) **Dataset.** A set of 366 webcam clips was collected, cut to a fixed length, filtered for tracking quality, labelled by hand, and processed with the same OpenFace/I3D pipeline as the public datasets. This test-only set measures performance on real students recorded on their own webcams, where camera angles and conditions differ from the public datasets (Section 3.2).

(iii) **Generalisation.** This thesis runs a 3×3 cross-dataset study (training on CMOSE, DAiSEE, or their combination; testing on each public test set plus the private set) showing that engagement models do not transfer to a new dataset, that accuracy hides this collapse while QWK exposes it, and that combined training is the simplest effective fix. On the private set the architecture and combined training reinforce each other: the proposed hybrid model trained on both datasets is the best model (QWK 0.379 against 0.285 for the best baseline, a wider gap than in-domain), which shows the value of the three contributions together (Sections A.5 and 4.7).

A reproducible pipeline turns about 1,500 trained runs into the figures and tables of this thesis, from feature extraction through training, cross-dataset evaluation, and figure and table generation.

## 1.5 Organization of Thesis

The remainder of this thesis is organised as follows.

Chapter 2 reviews vision-based engagement recognition and the background the thesis relies on. It sets the scope, compares the feature-based and end-to-end approaches with their limits, and then presents the background knowledge later chapters assume: the OpenFace and I3D feature extractors, the three temporal architecture families (TCN, LSTM, Transformer), the loss functions for imbalanced ordinal labels, the ordinal metrics, and the statistical measures used to analyse results.

Chapter 3 presents the methodology. It gives an overview of the pipeline with diagrams of the baseline and the proposed model, describes the three training datasets and their imbalance, the private test set, and the leak-free preprocessing, defines the single-encoder baselines, and builds the feature-group decomposition and the proposed hybrid model on top of them.

Chapter 4 reports the experimental results, centred on the proposed hybrid model. It sets out the loss forms, the training regime, and the $3\times3$ cross-dataset evaluation matrix; confirms that the evaluation protocol holds across the model's 486 configurations; answers the design questions (which encoder suits which group, and whether the I3D stream helps); and takes the model out of domain, where it outperforms the single-encoder baseline in every cell of the matrix and, with combined training, reaches the best result on the private set (RQ4).

Chapter 5 draws the findings together, restates the contributions in the light of the evidence, discusses the limits, and proposes future work: targeting the rarest disengaged class, explicit domain adaptation, and learned group encoders.

Appendix A presents the supporting single-encoder baseline study, protocol-first: it establishes the metrics (QWK, macro-accuracy, macro-MAE), the development dataset (CMOSE), and the development loss (cross-entropy), reads the in-domain leaderboard, studies prediction agreement to show the OpenFace and I3D families make different errors, and takes the baselines out of domain in the $3\times3$ matrix, setting the reference level the proposed model is measured against.

# CHAPTER 2. LITERATURE REVIEW

## 2.1 Scope of Research

This review is scoped to vision-based engagement recognition: classifying a learner's engagement from a frontal webcam recording of one person. Within this scope the thesis concentrates on four parts, around which the review is organised: (i) extracting facial-behaviour and motion features, (ii) modelling the per-frame descriptors in time, (iii) loss functions suited to imbalanced ordinal labels, and (iv) the metrics used to evaluate such models.

Several adjacent problems are left out of scope. Physiological sensors and interaction logs are not used, because they require instrumentation beyond a webcam. Facial-expression and emotion recognition are not treated as an end in themselves, and are referenced only where they inform how engagement features are designed. Finally, the thesis works with OpenFace and I3D features at the descriptor level rather than re-training the extractors: CMOSE ships these features precomputed, while for DAiSEE and the private set they are extracted from the raw video with the same toolkits (Section 3.2); how the extractors are trained internally is reviewed for understanding but not re-implemented.

## 2.2 Related Work

Engagement recognition rests on a small number of in-the-wild benchmarks. DAiSEE [2] is the most widely used: it provides short webcam clips annotated with four ordinal engagement levels under varied, uncontrolled conditions, which makes it realistic but noisy. CMOSE [1] is more recent and larger, also collected in the wild but annotated by experts, ships precomputed OpenFace and I3D features, and reports strong inter-rater reliability; its scale and label quality make it the thesis's primary dataset. A systematic review [8] surveys the common pipelines and names the two recurring open problems, severe class imbalance and subjective labels, which matches the motivation in Chapter 1.

One family of methods first reduces each frame to a compact descriptor with a toolkit such as OpenFace [3] and then models the sequence. Representative pipelines feed OpenFace action units, gaze, and head pose to a recurrent network, for example an OpenFace+BiLSTM model [4], or apply dimensionality reduction and resampling before a convolutional or fully-connected classifier [9]. Others aggregate the features over time and compare classical learners such as support vector machines and gradient-boosted trees [5]. These works show that OpenFace descriptors carry a real engagement signal, but they share one assumption: the

709-dimensional descriptor is treated as one uniform vector, encoded by one model with its temporal bias.

A second family learns directly from pixels. EfficientNetV2 [10] backbones followed by an LSTM [6] encode appearance per frame and integrate it over time, and context-aware affect-recognition models exploit spatial relationships in the scene [7]. These models capture appearance and motion cues that landmark-based descriptors discard, but they are heavier to train, need the raw video, and are less interpretable, because the learned representation is not decomposed into nameable behavioural channels.

Two gaps persist across both families and define the contribution of this thesis. First, the mixed facial features are passed through a single encoder rather than encoded per behavioural subsystem, even though their signals have very different timing. Second, studies rely on accuracy as the primary metric and lack a reliable evaluation of generalisation, since in-domain accuracy dominates the literature yet on imbalanced data a high accuracy can hide a model that does not transfer. The proposed hybrid model of Chapter 3 targets the first gap, and the $3\times3$ cross-dataset protocol of Chapter 4 targets the second.

## 2.3 Background Knowledge

This section presents the technical foundations the rest of the thesis depends on: the feature extractors (Section 2.3.1), the temporal architectures (Section 2.3.2), the loss functions (Section 2.3.3), the evaluation metrics (Section 2.3.4), and the statistical measures for result analysis (Section 2.3.5).

### 2.3.1 Feature Extraction

#### a, OpenFace 2.0 facial-behaviour descriptors

OpenFace 2.0 [3] is an open-source facial-behaviour analysis toolkit that processes a video frame by frame and, for each frame, estimates four kinds of measurement: the two- and three-dimensional facial landmarks that outline the face and eye regions, the head pose recovered from those landmarks, the eye-gaze direction vectors and angles for both eyes, and the intensity and presence of the facial action units of the Facial Action Coding System. Each frame is also tagged with a tracking-confidence and success flag that downstream pipelines use to discard poorly tracked frames.

In CMOSE the per-frame descriptor used here is 709-dimensional, and these numbers are not homogeneous. By the meaning of their columns they split into five feature groups, gaze (8 dimensions), eye landmarks (280), face landmarks (340), head pose (46), and action units (35), and the thesis exploits this split throughout. The groups are physically distinct subsystems with their own dynamics: gaze and eye landmarks change rapidly and noisily; head pose drifts slowly and smoothly; action units fire in short, sparse bursts. Section 3.4.1 formalises this decomposition and Section 3.4 turns it into an architecture.

#### b, I3D motion and appearance features

The Inflated 3D ConvNet (I3D) [11] is a video representation obtained by inflating the two-dimensional convolutional filters of an ImageNet-pretrained image network into three dimensions, so they operate over space and time jointly, and then pretraining on the large Kinetics dataset for action recognition. I3D therefore captures appearance and short-range motion that landmark-based descriptors omit. In the data used here, CMOSE ships a single precomputed 1024-dimensional I3D embedding per clip, and for DAiSEE and the private set the same vector is extracted from the raw video. In every case it is one vector pooled over the whole clip, not a per-frame sequence, so the thesis treats I3D as a global clip-level stream of motion and appearance that complements the OpenFace descriptors.

### 2.3.2 Temporal Architectures

A clip is a sequence of per-frame OpenFace descriptors, so a model must reduce a variable-length sequence to a single engagement prediction. Three architecture families dominate sequence modelling, and all three are used in this thesis, both as single-encoder baselines (Section 3.3) and as the per-group encoders inside the proposed hybrid model (Section 3.4). They differ in how they let one time step influence another.

#### a, Temporal Convolutional Networks (TCN)

A Temporal Convolutional Network [12] applies one-dimensional convolutions along the time axis, and two design choices make it well suited to behavioural sequences. First, the convolutions are causal: the output at time $t$ depends only on inputs at times $\leq t$, achieved by padding on the left and removing the surplus on the right. Second, successive layers use exponentially increasing dilation, so a layer at depth $l$ skips $2^l$ steps and a stack of $L$ layers reaches a receptive field that grows as $\mathcal{O}(2^L)$ with few parameters. Each layer is a residual block (two dilated convolutions with weight normalisation, a rectified-linear nonlinearity, and dropout, summed with a possibly projected copy of the input) that stabilises deep stacks (Figure 2.1). The result is a strong inductive bias for short, local temporal motifs at multiple scales, which matches signals such as a head nod or an action-unit burst.

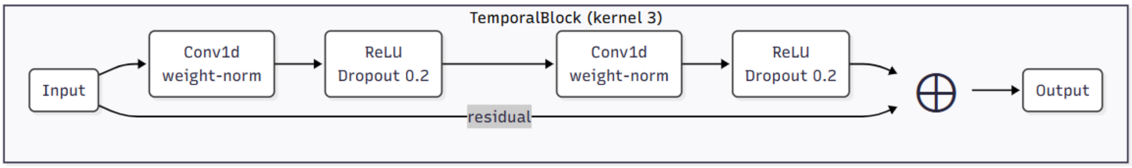



**Figure 2.1:** Residual TemporalBlock

#### b, Long Short-Term Memory (LSTM)

The Long Short-Term Memory network [13] is a recurrent architecture that processes the sequence one step at a time while maintaining a cell state $c_t$ and a hidden state $h_t$ (Figure 2.2). Three gates (input $i_t$, forget $f_t$, and output $o_t$) regulate how much new information is written, how much past information is retained, and how much of the cell state is exposed:

$$\begin{aligned} f_t &= \sigma(W_f x_t + U_f h_{t-1} + b_f), & i_t &= \sigma(W_i x_t + U_i h_{t-1} + b_i), \\ o_t &= \sigma(W_o x_t + U_o h_{t-1} + b_o), & \tilde{c}_t &= \tanh(W_c x_t + U_c h_{t-1} + b_c), \\ c_t &= f_t \odot c_{t-1} + i_t \odot \tilde{c}_t, & h_t &= o_t \odot \tanh(c_t), \end{aligned} \tag{2.1}$$

where $\sigma$ is the logistic sigmoid and $\odot$ is the element-wise product. The gating lets the network carry information across many time steps without the vanishing-gradient problem of plain recurrent networks, giving it a bias toward longer-range dependencies. The cost is that the sequential recurrence is harder to parallelise than convolution or attention.

#### c, Transformers

The Transformer encoder [14] drops recurrence and convolution and instead relates all time steps directly through self-attention. For queries $Q$, keys $K$, and values $V$ obtained by linear projections of the input, attention computes

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right) V, \tag{2.2}$$

so every output position is a learned weighted combination of every input position, with $d_k$ the key dimension. Because attention is permutation-invariant, order is injected through a positional encoding added to the input; this thesis uses the standard sinusoidal encoding. Multiple attention heads run in parallel and their outputs are concatenated, and each attention sub-layer is followed by a position-wise feed-forward network, with residual connections and layer normalisation throughout (Figure 2.3). The Transformer has the weakest temporal prior of the three families, since it can attend anywhere, which makes it flexible but more reliant on data.

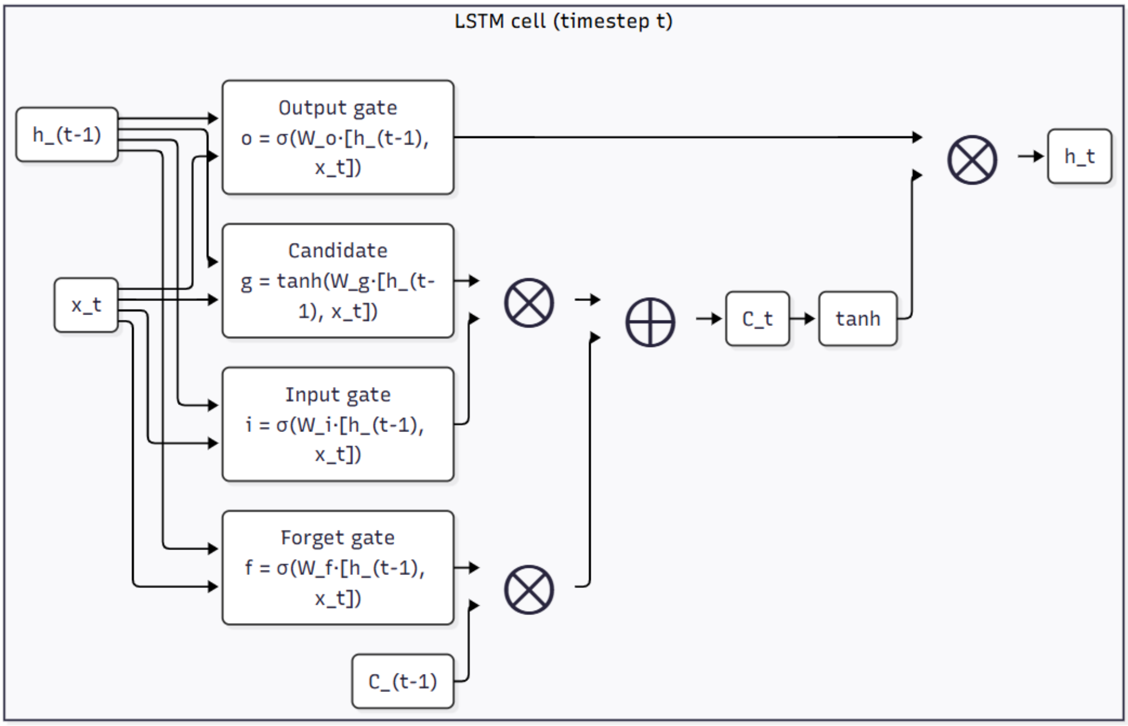



**Figure 2.2:** LSTM cell at time step $t$

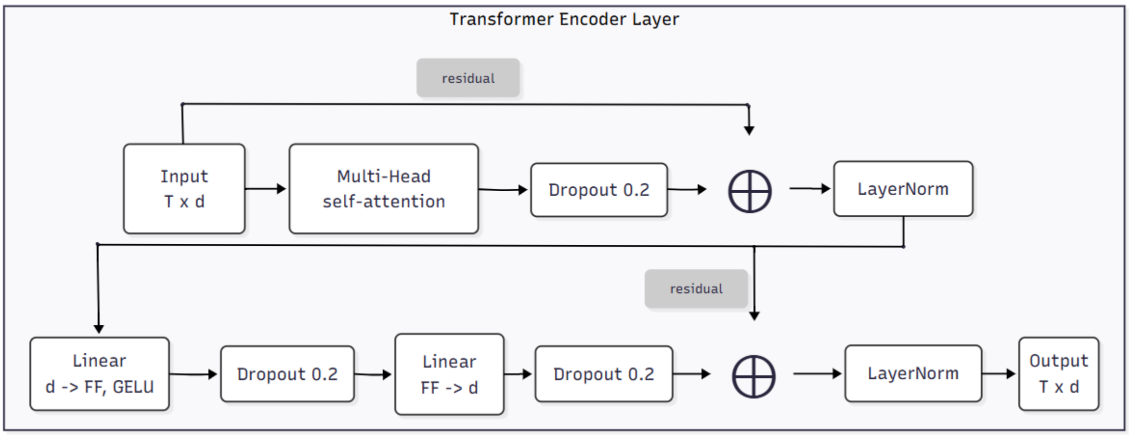



**Figure 2.3:** Transformer encoder layer

### 2.3.3 Loss Functions for Imbalance and Ordinality

Let a model produce logits $z \in \mathbb{R}^C$ for the $C = 4$ engagement classes. The softmax function maps these logits to a probability distribution over the classes, the predicted probability of class $c$ being

$$\widehat{y}_c = \mathrm{softmax}(z)_c = \frac{e^{z_c}}{\sum_{j=1}^{C} e^{z_j}} \tag{2.3}$$

so that $\widehat{y}_c \in (0,1)$ and $\sum_{c=1}^{C} \widehat{y}_c = 1$. Let $y \in \{1,\dots,C\}$ be the ordinal ground-truth class, encoded as a one-hot vector through the indicator function $\mathbf{1}[\,\cdot\,]$, which returns $1$ when its argument is true and $0$ otherwise:

$$\mathbf{1}[\,S\,] = \begin{cases} 1 & \text{if the statement } S \text{ is true} \\ 0 & \text{if } S \text{ is false} \end{cases} \tag{2.4}$$

so that the one-hot component of the label is $y_c = \mathbf{1}[\,c = y\,]$. Three loss families are compared in this thesis; their exact application is given in Section 4.2.1.

**Cross-entropy (CE)** is the standard classification loss,

$$\mathcal{L}_{\text{CE}} = -\sum_{c=1}^{C} y_c \log \widehat{y}_c, \tag{2.5}$$

which, because $y$ is one-hot, reduces to the negative log-probability of the true class; it treats all classes and all errors as equally severe. On imbalanced data it is dominated by the majority class and on ordinal data it ignores class order, so it serves as the baseline against which the other losses are judged.

**Weighted cross-entropy** counteracts imbalance by scaling each class's contribution by a weight $w_c$ inversely proportional to its frequency,

$$\mathcal{L}_{\text{WCE}} = -\sum_{c=1}^{C} w_c\, y_c \log \widehat{y}_c. \tag{2.6}$$

This raises the cost of mistakes on rare classes such as E0 (Highly-Disengage) and stops the model collapsing onto the majority, at the expense of some accuracy. Resampling methods such as SMOTE [15] address the same imbalance at the data level; this thesis instead studies loss-level remedies, which require no change to the data pipeline.

**Ordinal (squared-EMD) loss** [16] encodes the order of the classes by comparing cumulative distributions. Writing the predicted cumulative distribution as $\sum_{j=1}^{k} \widehat{y}_j$ and that of the one-hot label as the unit step $\sum_{j=1}^{k} y_j$, the loss is the mean squared distance between the two step functions,

$$\mathcal{L}_{\text{EMD}} = \frac{w_y}{C} \sum_{k=1}^{C} \left( \sum_{j=1}^{k} \widehat{y}_j - \sum_{j=1}^{k} y_j \right)^2 , \tag{2.7}$$

the squared Earth Mover's Distance (also called the CDF-MSE) between the predicted and target distributions over the ordered classes [16], not the unsquared 1-Wasserstein distance; the leading $w_y$ is the weight of the true class (Section 4.2.1), set to $1$ in the unweighted variant. Because moving probability mass across several classes costs more than moving it to an adjacent class, this loss directly penalises the large ordinal errors (E0↔E3) that accuracy and plain cross-entropy treat as ordinary.

All models are optimised with Adam [17], an adaptive first-order optimiser that maintains per-parameter running estimates of the first and second moments of the gradient.

### 2.3.4 Evaluation Metrics

Because the labels are ordinal and imbalanced, the choice of metric is itself a research question (RQ4). This thesis reports six metrics over a test set of $N$ samples with true labels $y_i$ and predicted labels $\widehat{y}_i$, but treats exactly one, QWK, as the primary metric that selects the best model; the other five are secondary, reported to describe a result but never to choose one. Where a confusion matrix is needed, $O_{ij}$ denotes the number of samples of true class $i$ predicted as class $j$.

**Accuracy** is the fraction of correct predictions,

$$\text{Accuracy} = \frac{1}{N} \sum_{i=1}^{N} \mathbf{1}[\, y_i = \widehat{y}_i \,] . \tag{2.8}$$

It is reported for comparability with the literature but is not a primary metric here, because on imbalanced data it rewards majority prediction.

**Macro-accuracy** (balanced accuracy) is the mean of the per-class recalls,

$$\text{MacroAcc} = \frac{1}{C} \sum_{c=1}^{C} \frac{\sum_{i=1}^{N} \mathbf{1}[\, y_i = \widehat{y}_i = c \,]}{\sum_{i=1}^{N} \mathbf{1}[\, y_i = c \,]}, \tag{2.9}$$

giving every class equal weight regardless of frequency. It exposes the minority-class collapse that accuracy hides; here it is a secondary metric, reported to characterise class balance but never used to select a model.

**Mean absolute error (MAE)** treats the class indices as an ordinal scale and averages the distance between truth and prediction,

$$\mathrm{MAE} = \frac{1}{N}\sum_{i=1}^{N} |y_i - \widehat{y}_i| \,, \tag{2.10}$$

while macro-MAE averages the per-class MAE so errors on rare classes are not drowned out,

$$\mathrm{MacroMAE} = \frac{1}{C}\sum_{c=1}^{C} \frac{\sum_{i=1}^{N} \mathbf{1}[\, y_i = c\,]\, |y_i - \widehat{y}_i|}{\sum_{i=1}^{N} \mathbf{1}[\, y_i = c\,]}. \tag{2.11}$$

Both reward predictions that are close on the scale even when not exactly right, and both are smaller-is-better. Macro-MAE is the natural ordinal-distance companion to QWK, with each class weighted equally so the rare disengaged classes are not masked; like the other secondary metrics it is reported alongside QWK and never used to select a model.

**Cohen's kappa** [18] measures agreement corrected for chance. With observed agreement $p_o$ and expected agreement $p_e$, $\kappa = (p_o - p_e)/(1 - p_e)$. In its weighted form it is

$$\kappa = 1 - \frac{\sum_{ij} w_{ij}\, O_{ij}}{\sum_{ij} w_{ij}\, E_{ij}}, \tag{2.12}$$

where $E_{ij}$ is the expected count under independent marginals and $w_{ij}$ is a penalty matrix. The unweighted variant uses $w_{ij} = 1$ off the diagonal and $0$ on it.

**Quadratic-weighted kappa (QWK)** is the same coefficient with the quadratic penalty

$$w_{ij} = \frac{(i-j)^2}{(C-1)^2}, \tag{2.13}$$

so disagreements are penalised in proportion to the square of the ordinal distance between the true and predicted classes. QWK suits this problem because it is corrected for chance, respects class order, and cannot be inflated by majority prediction. It is the thesis's primary metric and the only criterion used to select the best model, with macro-accuracy and macro-MAE reported as secondary companions that describe a result but never choose it.

### 2.3.5 Statistical Measures for Result Analysis

Beyond scoring each model, the results chapters ask three comparative questions that need their own tools: do the six metrics rank the configurations the same way (Section A.2.1), does in-domain strength predict cross-dataset strength (Section A.5.3), and do different models make the same per-clip decisions (Section A.4)? The first two are rank-correlation questions; the third is an agreement question.

**Spearman's rank correlation** $\rho$ [19] measures how well a monotonic relationship describes two variables by applying the Pearson correlation to their ranks. For $n$ paired observations with rank differences $d_i$ and no ties it reduces to

$$\rho = 1 - \frac{6\sum_{i=1}^{n} d_i^2}{n\,(n^2-1)}, \tag{2.14}$$

ranging from $-1$ (perfectly inverted) through $0$ (no monotonic relation) to $+1$ (identical ordering). This thesis uses $\rho$ to test whether a model's in-domain QWK predicts its QWK on unseen target datasets.

**Kendall's** $\tau$ [20] measures rank agreement through concordance. Counting the pairs ordered the same way in both variables (concordant, $n_c$) against those ordered oppositely (discordant, $n_d$), with no ties,

$$\tau_a = \frac{n_c - n_d}{\binom{n}{2}}, \tag{2.15}$$

again bounded by $[-1, 1]$. It is less sensitive to small samples and to a few large rank displacements than $\rho$, so it is used here to quantify how far the six metrics agree on the ranking of the baseline configurations.

Prediction agreement asks whether two models agree clip by clip rather than merely score alike, measured by Cohen's $\kappa$ (Section 2.3.4) between the two models' predicted-label vectors; chance-correction matters because two models that lean toward the majority class agree often by default. The headroom from combining a pool of $M$ configurations is bounded by the oracle accuracy, the fraction of clips at least one configuration classifies correctly,

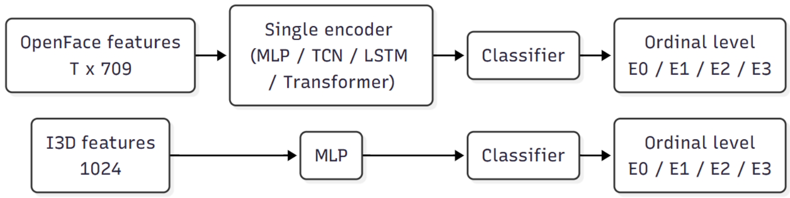
$$\mathrm{OracleAcc} = \frac{1}{N}\sum_{n=1}^{N} \mathbf{1}\left[\exists\, m \in \{1,\dots,M\} :\ \widehat{y}_n^{(m)} = y_n\right], \tag{2.16}$$

which upper-bounds any fixed combiner and is contrasted with the realistic majority-vote accuracy. A large gap between the best single model, the majority vote, and the oracle indicates that the models err on different clips and that an ensemble has room to improve.

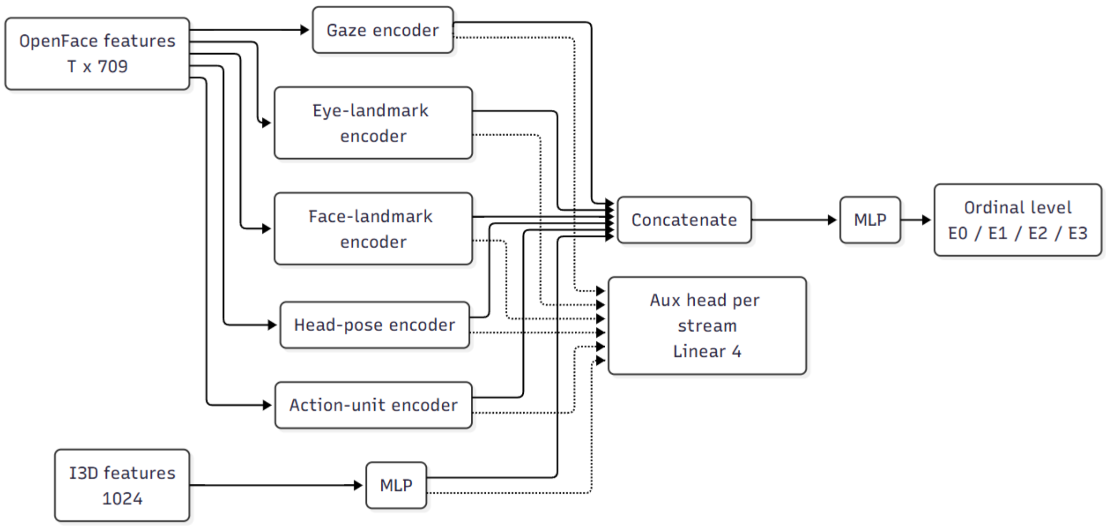
## 2.4 Chapter Summary

This chapter scoped the thesis to ordinal engagement recognition from a webcam and analysed the two main approaches in the literature, identifying their shared limits: mixed facial descriptors are encoded as one block, and transfer across datasets is rarely measured with metrics that resist majority-class inflation. It then laid the background knowledge the rest of the thesis depends on: the OpenFace 709-dimensional descriptor and its five feature groups, the clip-level I3D embedding, the three temporal architecture families (TCN, LSTM, Transformer) and their different temporal biases, the three loss families for imbalance and ordinality, the six metrics with QWK primary and macro-accuracy and macro-MAE secondary, and the rank-correlation and prediction-agreement statistics used to analyse the results. Chapter 3 combines these foundations into a concrete methodology: it turns the five-group decomposition into the proposed hybrid model and specifies the single-encoder baselines it is compared against.

## 3.1 Overview




**(a)** Single-encoder baselines.




**(b)** Proposed feature-group hybrid model (detailed in Figure 3.4).

**Figure 3.1:** The baseline and the proposed hybrid model.

Two public datasets form the training data, while a self-collected set recorded on real webcams serves as an in-the-wild test set (Section 3.2). Every clip carries one of four ordinal engagement levels, from highly disengaged to highly engaged. Each clip is represented by per-frame OpenFace descriptors and a clip-level I3D embedding extracted from the raw video, then resampled to a common length and standardised with training-set statistics so that no target information leaks into preprocessing.

Two model paradigms are compared, contrasted in Figure 3.1. The single-encoder baselines follow the standard approach, in which one encoder, an MLP, TCN, LSTM, or Transformer, consumes the full 709-dimensional OpenFace sequence, or an MLP consumes the pooled I3D vector, before a classifier produces the prediction (Section 3.3).

The proposed hybrid model replaces that single encoder with a split-and-encode design: the OpenFace descriptor is split into five meaning-based feature groups, each encoded independently by its own temporal encoder, the I3D vector is encoded by an MLP, and the resulting embeddings are concatenated and classified by a shared MLP while an auxiliary head supervises every stream (Section 3.4). The loss functions, training regime, and 3×3 cross-dataset protocol under which every model is trained and evaluated are established with the experiments that use them in Chapter 4.

## 3.2 Data and Preprocessing

### 3.2.1 Public datasets

Two public datasets supply the labelled training data. DAiSEE [2] is a widely used in-the-wild engagement dataset of short frontal webcam clips, each showing a single learner under uncontrolled conditions and annotated on a four-level ordinal engagement scale, which makes it realistic but noisy; it ships official Train, Validation, and Test partitions used here directly for training, early-stopping validation, and testing. CMOSE [1] is a more recent and larger dataset, likewise collected in the wild but annotated by experts and reporting strong inter-rater agreement, and it ships precomputed OpenFace and I3D features; it is divided into a labelled training split, an `unlabel` split repurposed as the validation set for early stopping and checkpoint selection, and a test split. Both label every clip on the same four ordinal levels used throughout the thesis, E0 (Highly-Disengage), E1 (Disengage), E2 (Engage), and E3 (Highly-Engage), so that CMOSE, with its greater scale and label quality, serves as the primary source and DAiSEE as an independent second source.

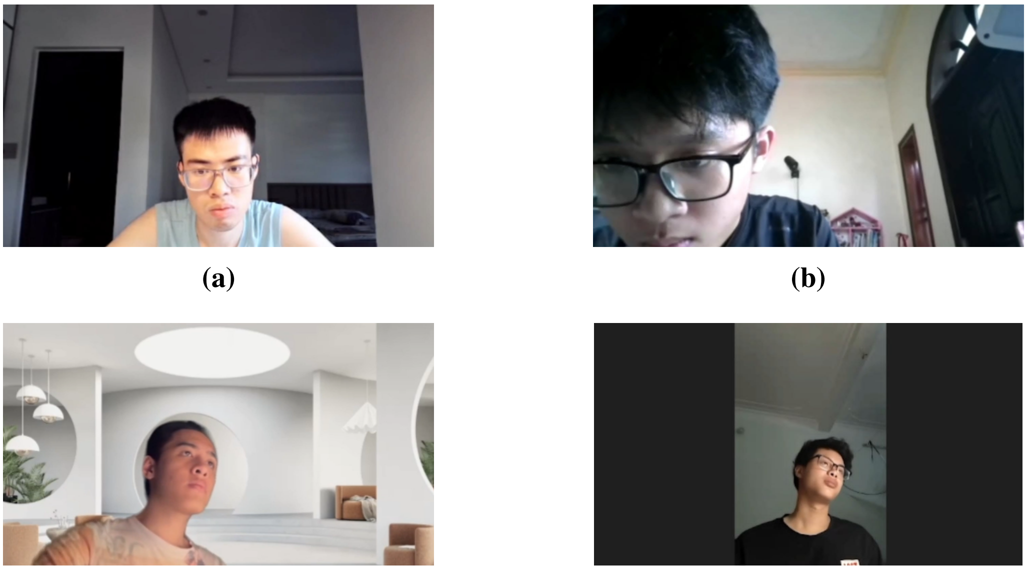
**Table 3.1:** Dataset statistics and class balance (%).

| Dataset | Total | Train | Val | Test | E0% | E1% | E2% | E3% |
|---|---|---|---|---|---|---|---|---|
| CMOSE | 12197 | 8783 | 2193 | 1221 | 2.800 | 18.100 | 69.400 | 9.600 |
| DAiSEE | 8571 | 5358 | 1429 | 1784 | 0.700 | 5.100 | 50.300 | 43.800 |

Table 3.1 shows that both datasets are heavily imbalanced toward the upper-middle engagement levels, and imbalanced in different directions: CMOSE concentrates on the single Engage level, whereas DAiSEE divides more evenly between Engage and Highly-Engage. The disengaged levels are rare in both, yet they are precisely the responses a monitoring system must catch. The same skew holds within each individual split, so it reflects the nature of the task rather than an artefact of how either dataset was partitioned.

### 3.2.2 Private dataset

The private set exists to measure deployment behaviour rather than in-dataset behaviour. No model is trained or selected on it, so it is a test-only set, and it is sampled from the webcam recordings a deployed model would actually face, so it probes a condition the public datasets leave untested. The reason this matters is geometric: the OpenFace gaze, head-pose, and landmark features are read from the camera's view of the face, so the same behaviour recorded from a different camera angle produces different feature values. Figure 3.2 shows private clips that differ along several axes: the camera's height and distance relative to the student's eye level, its position relative to the screen, its orientation, whether vertical or horizontal, and the background behind the learner, whether the real room, blurred, or a green screen. CMOSE and DAiSEE were recorded under far more uniform setups and so cannot exercise this variation, which makes a small, locally collected, hand-labelled set drawn from realistic online-class recordings the most direct way to check that a model stays fit once the camera geometry changes.



(a) (b) (c) (d)

**Figure 3.2:** Representative frames from the private set, illustrating the range of camera angles found in unconstrained online-class recordings.

The set was built with the same feature pipeline as the public data. Webcam recordings of online-class sessions were sampled at several points and cut into fixed-length ten-second clips, each containing a single learner, and every clip was run through the OpenFace pipeline used for the public datasets, with clips that retained too few tracked frames discarded as unreliable. The accepted clips were then labelled through a purpose-built local interface, one engagement level per clip and by a single annotator, a reliance revisited as a limitation in Section 5.4. Finally, the same OpenFace (709-d per frame) and I3D (1024-d per clip) extractors used for DAiSEE were applied, so the private features share the format of the precomputed CMOSE features and are distributionally comparable to the training data, and the 366 clips with complete, aligned OpenFace and I3D features form the final set. Its label prior differs again from both public datasets and, together with its independent recording conditions, makes it an in-the-wild measure of deployment behaviour that anchors Section 4.7.

### 3.2.3 Preprocessing

Each clip is represented by two feature streams. OpenFace 2.0 [3] produces a 709-dimensional descriptor for every frame, and I3D [11] produces a single 1024-dimensional embedding pooled over the whole clip. CMOSE ships both precomputed; for DAiSEE and the private set they are extracted from the raw video with the same toolkits, so all three sources share one feature format. When an OpenFace frame reports more than one detected face the highest-confidence row is kept, and any non-numeric entry is set to zero. Because clips differ in length, each OpenFace sequence is resampled along time to a fixed number of frames by per-feature linear interpolation, 300 frames by default, reduced to 150 for the largest training runs to fit memory and to 75 in the multimodal variant, while the I3D stream, being a single vector per clip, needs no temporal resampling and is encoded by an MLP (Section 3.4).

Every feature is then standardised by a per-feature $z$-score. The mean and standard deviation are fit on the training split alone, and the validation, test, and private sets are transformed with those training statistics, so no target-set information leaks into preprocessing. The OpenFace and I3D streams are standardised with separate statistics.

## 3.3 Baseline Models

Five single-encoder baselines provide the comparison points and form the allow-list `MODEL_ORDER` used by the analysis. Each applies one encoder to the full feature sequence before a classifier.

(i) `openface_mlp`: flattens the OpenFace sequence and classifies it with an MLP (→256→128→4, dropout 0.3).

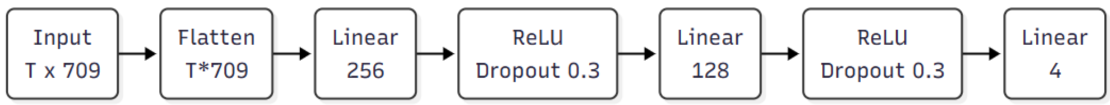
(ii) `openface_tcn`: a TCN over the 709-d sequence (residual dilated blocks of widths $256, 128, 128$; kernel 3; dilations $1, 2, 4$), adaptive average-pooling, then a $128 \to 128 \to 4$ classifier.

(iii) `openface_lstm`: a two-layer LSTM (hidden width 256) whose final hidden state feeds a $256 \to 128 \to 4$ classifier.

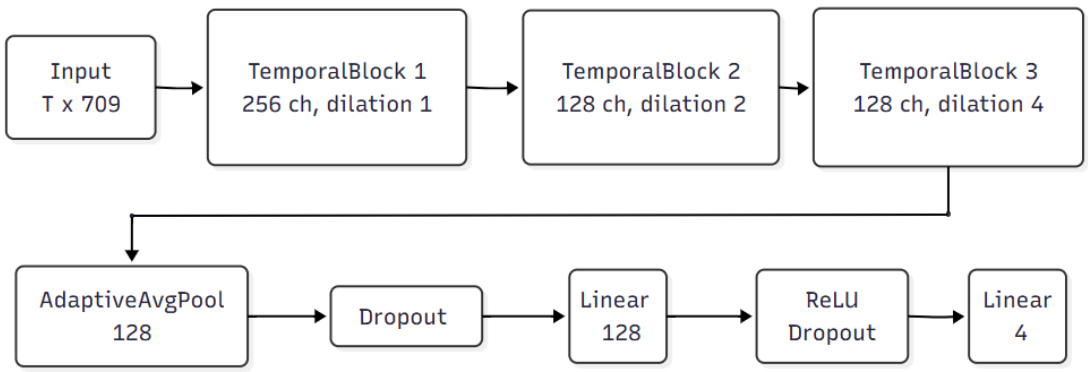
(iv) `openface_transformer`: a linear projection to 128 dimensions, a sinusoidal positional encoding, and two Transformer encoder layers (4 heads, feed-forward width 256), mean-pooled and classified.

(v) `i3d_mlp`: classifies the single 1024-d I3D clip vector with an MLP ($1024 \to 256 \to 128 \to 4$, dropout 0.3).

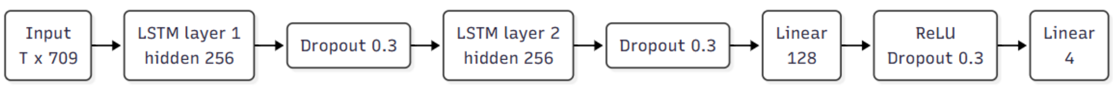
The five baseline architectures are shown in Figure 3.3. The four OpenFace baselines (a–d) encode the full $T \times 709$ sequence with a single encoder, while `i3d_mlp` (e) classifies the pooled 1024-d I3D clip vector. Forcing one encoder, and therefore one temporal bias, onto all 709 mixed features is the design decision the proposed model revisits in Section 3.4.




**(a)** `openface_mlp`.

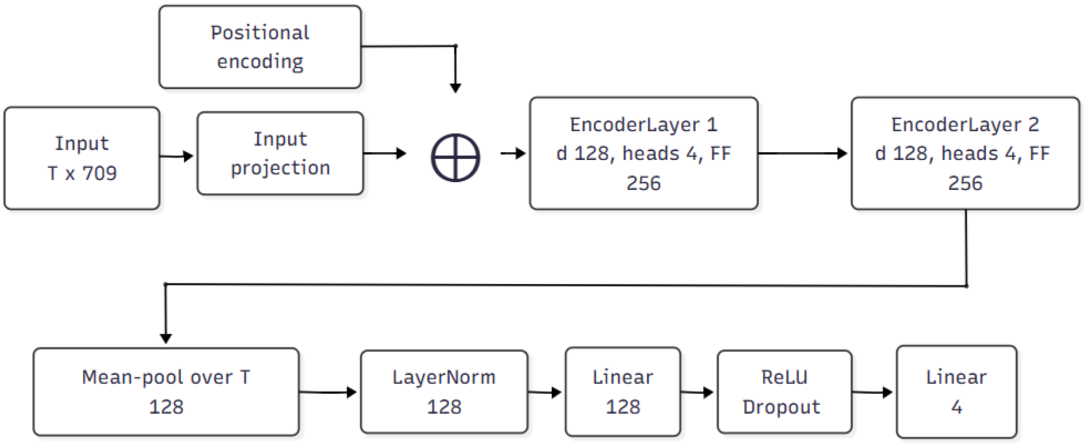



**(b)** `openface_tcn`.




**(c)** `openface_lstm`.

**Figure 3.3:** The five single-encoder baselines.




**(d)** `openface_transformer`.


**(e)** `i3d_mlp`.

**Figure 3.3:** The five single-encoder baselines (continued).

## 3.4 Proposed Hybrid Model

### 3.4.1 Grouping the facial features by meaning

The 709 OpenFace features are partitioned into five meaningful groups by matching the column names, in the fixed order used everywhere in the pipeline (gaze, eye, face, head, action units). Table 3.2 summarises the partition; every one of the 709 columns belongs to exactly one group.

**Table 3.2:** Grouping of the 709-dimensional OpenFace descriptor into five feature groups by meaning.

| Group | Content | Dim |
|---|---|---|
| Gaze | gaze direction vectors and angles | 8 |
| Eye landmarks | 2-D/3-D eye-region landmark coordinates | 280 |
| Face landmarks | 2-D/3-D facial landmark coordinates | 340 |
| Head pose | head translation and rotation parameters | 46 |
| Action units | facial action-unit presence and intensity | 35 |
| **Total** | | **709** |

This decomposition is the inductive prior of the method. The five subsystems have different dynamics, so each can be matched to an encoder whose temporal bias suits it: gaze and eye landmarks show rapid, noisy micro-saccades; head pose drifts slowly and smoothly; action units fire in brief, sparse bursts. Encoding all 709 features together, as the single-encoder baselines of Section 3.3 do, forces a single bias on all of them at once.

### 3.4.2 Model architecture

The proposed hybrid model, shown in Figure 3.4, encodes each stream independently, fuses the resulting embeddings, and supervises every stream with its own auxiliary head. It is built from the same three encoder families as the OpenFace baselines of Section 3.3 (TCN, Transformer, LSTM), but where a baseline applies one encoder to the full $T \times 709$ sequence, the proposed model assigns an independently chosen encoder to each of the five feature groups defined above, so any difference in performance is attributable to the decomposition rather than to a different kind of encoder.

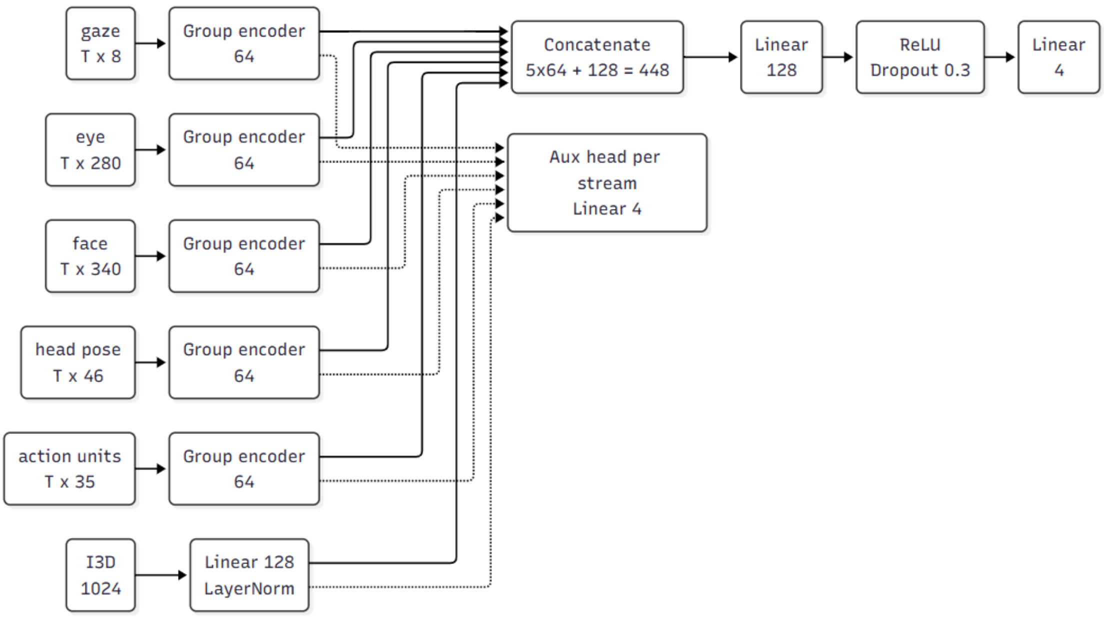


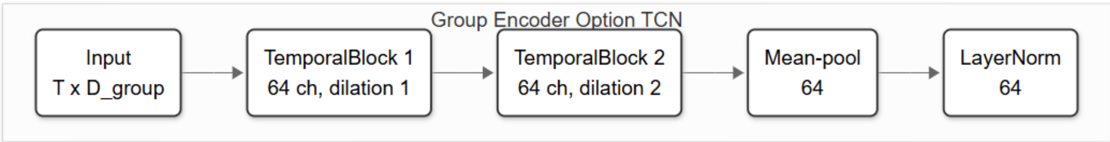

**Figure 3.4:** Proposed hybrid model architecture (I3D stream enabled; the I3D-stream-disabled variant drops the I3D row).

Each group is encoded by a `GroupEncoder`, one of three interchangeable temporal encoders (Figure 3.5), followed by temporal mean-pooling and a layer normalisation, producing a 64-dimensional embedding:

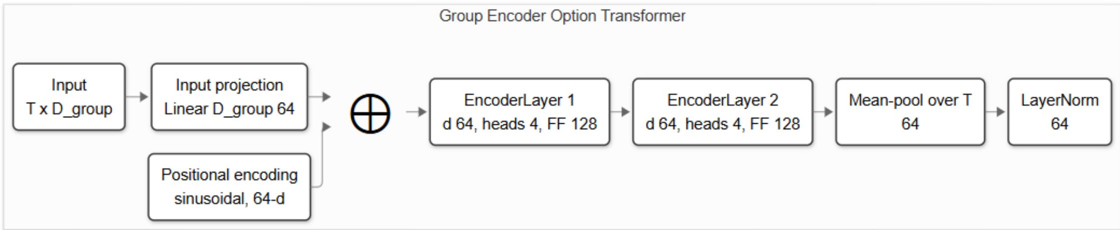
(i) a TCN: two residual blocks of dilated causal 1-D convolutions (channel widths $64, 64$; kernel size 3; dropout 0.2);

(ii) a Transformer encoder: a linear projection to 64 dimensions, a sinusoidal positional encoding, and two encoder layers (4 heads, feed-forward width 128, GELU, dropout 0.2);

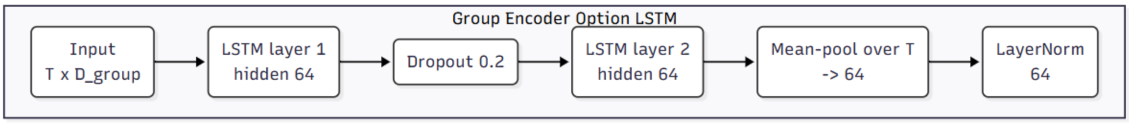
(iii) an LSTM: a two-layer recurrent encoder of hidden width 64 (dropout 0.2).




**(a)** TCN group encoder.




**(b)** Transformer group encoder.




**(c)** LSTM group encoder.

**Figure 3.5:** The three interchangeable per-group encoders.

The five 64-d group embeddings are concatenated into a 320-d vector and passed through a two-layer MLP classifier ($320 \to 128 \to 4$, ReLU, dropout 0.3) that produces the main logits. Each group additionally feeds an auxiliary classification head ($\text{dropout} \to \text{Linear}(64 \to 4)$). The total loss is the main-head loss plus `aux_weight` $= 0.2$ times the mean of the per-group losses (Section 4.2.1); the auxiliary supervision makes each stream individually discriminative.

The I3D-stream-enabled variant adds a sixth stream: the single 1024-d I3D clip vector is encoded by an MLP ($1024 \to 256 \to 128$, ReLU, dropout 0.3) and layer-normalised to a 128-d embedding with its own auxiliary head, so the fusion MLP takes a 448-d input ($5 \times 64 + 128$). The I3D stream uses an MLP rather than a temporal encoder because the provided I3D feature is a single pooled vector, not a sequence.

A configuration of the proposed hybrid model is fully specified by the encoder chosen for each of the five groups, written as an `arch_key` of five tokens in group order, for example `TCN_T_TCN_LSTM_T` (`T` = Transformer, `TCN` = TCN, `LSTM` = LSTM). With three choices per group there are $3^5 = 243$ configurations, each evaluated with the I3D stream enabled and disabled (Section 4.3).

## 3.5 Chapter Summary

This chapter specified the methodology end to end: the pipeline and its two model paradigms (Figure 3.1); the three training datasets and their imbalance, the test-only private set and its collection, and the leak-free preprocessing; the five single-encoder baselines; the five-group decomposition of the OpenFace descriptor; and the proposed hybrid model (Figure 3.4) that reuses the baselines' encoder families per group, with its fusion and auxiliary heads, I3D variant, and 243-configuration design space. The experimental apparatus that applies this methodology, its loss forms, training regime, and $3 \times 3$ evaluation protocol with its single primary metric (QWK) and secondary metrics, is set out in the experimental-results chapter (Chapter 4); the supporting single-encoder baseline study is given in Appendix A.

## 4.1 Overview

This chapter reports the experimental results, centred on the proposed hybrid model of Section 3.4. Two questions are put to it: whether its best configuration improves on the single-encoder reference in-domain, and whether the architecture and dataset combination reinforce each other out of domain, where deployment actually happens.

Two findings from the single-encoder baseline study frame the comparison and are established in Appendix A. First, the strongest baseline reaches QWK 0.537, the reference level the proposed model must exceed. Second, the OpenFace and I3D feature families, though they score almost the same, fail on different clips, so a model that learns to combine complementary streams should improve on any single encoder. That study also fixes the two protocol choices this chapter inherits, cross-entropy as the development loss and QWK as the single selection metric, and justifies them empirically.

The chapter proceeds as a sequence of experiments, each stating its aim, its setting, its result, and the conclusion that leads to the next. Section 4.2 states how the experiments are conducted, including the loss forms, the training regime, and the 3×3 evaluation matrix that every result in the thesis uses.

## 4.2 Experimental Setup

The experiments evaluate one architecture family, the proposed hybrid model of Section 3.4, over its full design space. All $3^5 = 243$ per-group encoder assignments are instantiated, each with the I3D stream enabled and disabled, giving 486 configurations. Every configuration is trained with cross-entropy, the development loss established in the baseline study (Appendix A, Section A.2.3), on each of the three sources and evaluated under the training regime and 3×3 protocol defined below. Fixing the loss isolates the effect of the architecture.

The reporting follows one metric policy throughout: QWK selects every best model, macro-accuracy and macro-MAE accompany the headline results, and accuracy appears only as the cautionary metric, for the last time in Section 4.7.1. The empirical case for this policy is made on the baselines in Appendix A. A headline comparison reports the tuned maximum of QWK over the design space; a claim about the family is reported as a mean and a distribution over the 243 or 486 configurations, so a family-level effect is robustness across the design space rather than a single tuned point.

The experiments follow the four research questions. Section 4.3 confirms that the evaluation protocol carries over to this larger population (RQ1's frame condition); Section 4.4 isolates which group's encoder choice matters (RQ2); Section 4.5 measures I3D fusion with a paired ablation (RQ3); Section 4.6 pins down the in-domain comparison with the baseline (RQ1); Section 4.7 takes the model out of domain and to the private set (RQ4); and Section 4.8 discusses why the design works.

### 4.2.1 Loss Functions

Three loss families are used (Section 2.3.3): cross-entropy, weighted cross-entropy, and the ordinal squared-EMD loss of Equation (2.7). The baseline study sweeps all three to measure the effect of the loss (Appendix A); the proposed model is trained only with the development loss selected there. The class weights for the weighted and ordinal variants are inverse-frequency weights computed on the training split and normalised to mean 1: for class $c$ with $n_c$ training samples and $C$ non-empty classes,

$$w_c = \frac{1}{n_c} \cdot \frac{C}{\sum_k \frac{1}{n_k}}, \qquad \text{so that} \quad \frac{1}{C} \sum_c w_c = 1. \tag{4.1}$$

The ordinal loss is applied in its class-weighted form as well, multiplying each sample's CDF distance by $w_y$.

For the proposed hybrid model the objective also includes the auxiliary term. With main logits $z$ and per-stream auxiliary logits $\{a^{(s)}\}$, the total loss is

$$\mathcal{L} = \mathcal{L}_{\text{main}}(z, y) \; + \; \lambda_{\text{aux}} \cdot \frac{1}{S} \sum_{s=1}^{S} \mathcal{L}_{\text{CE}}\big(a^{(s)}, y\big), \tag{4.2}$$

where $\mathcal{L}_{\text{main}}$ is the main-head loss, $\lambda_{\text{aux}} = 0.2$ is the auxiliary weight, and the auxiliary heads are always trained with plain cross-entropy over the $S$ streams (five groups, or six with the I3D stream enabled). The formulation admits any of the three losses for the main head; the ablation here fixes $\mathcal{L}_{\text{main}}$ to cross-entropy, so its results isolate the effect of the architecture from that of the loss.

### 4.2.2 Training Protocol

All models are optimised with Adam [17] at a learning rate of $1 \times 10^{-4}$. Training uses mini-batches (128 for the baselines, 64 for the ablation) and early stopping on the validation loss (the CMOSE `unlabel` split and its analogues) with patience 10 and an epoch cap (800 for baselines, 400 for the ablation). The checkpoint with the lowest validation loss is restored for testing. Following the convention adopted in this project, only the training and validation loss are recorded per epoch; the final metrics are evaluated once, on the held-out test split. Runs use a fixed seed (42) and optional mixed precision on GPU, and every run converges stably and early-stops well before the cap, so the reported differences are architectural rather than optimisation artefacts.

### 4.2.3 Metrics and the 3×3 Evaluation Matrix

Generalisation is measured with a 3×3 matrix: every model and loss is trained on each of {CMOSE, DAiSEE, Combined} and evaluated on each of {CMOSE-test, DAiSEE-test, Private}. The diagonal cells are the in-domain results; the off-diagonal cells and the Private column measure transfer to unseen distributions.

Each evaluation reports the six metrics of Section 2.3.4 (accuracy, macro-accuracy, MAE, macro-MAE, Cohen's kappa, and QWK), but QWK is the single primary metric: it alone selects the best model. The other five are secondary, reported for analysis and comparability but never used to pick a winner. The empirical justification for selecting on QWK alone is given in the baseline study (Appendix A, Section A.2). Row-normalised confusion matrices and per-class F1 scores are used for error analysis.

The study comprises roughly 1,500 trained runs (the five baselines × three losses × three sources, and the 243 per-group configurations of the proposed hybrid model with the I3D stream enabled and disabled); all are aggregated programmatically, and every figure and table of this chapter and Appendix A is produced from the aggregate by a single command.

## 4.3 Evaluation Properties of the Proposed Model

The evaluation protocol of Section 4.2.3 was fixed on fifteen single-encoder baselines (Appendix A); this experiment confirms that its three properties hold for 486 configurations of a different architecture family, none of which existed when the protocol was set. Figure 4.1 plots the distribution of the two variants of the proposed hybrid model (I3D stream enabled and disabled) on every metric, each panel against the best baseline on that metric, so all six metrics appear here for reference. Three properties reproduce.

The metric structure reproduces. The Kendall rank correlation between the six metrics over the 486 in-domain configurations gives the same two-block picture as the baseline study (Appendix A, Section A.2.1), a micro block, a macro block,

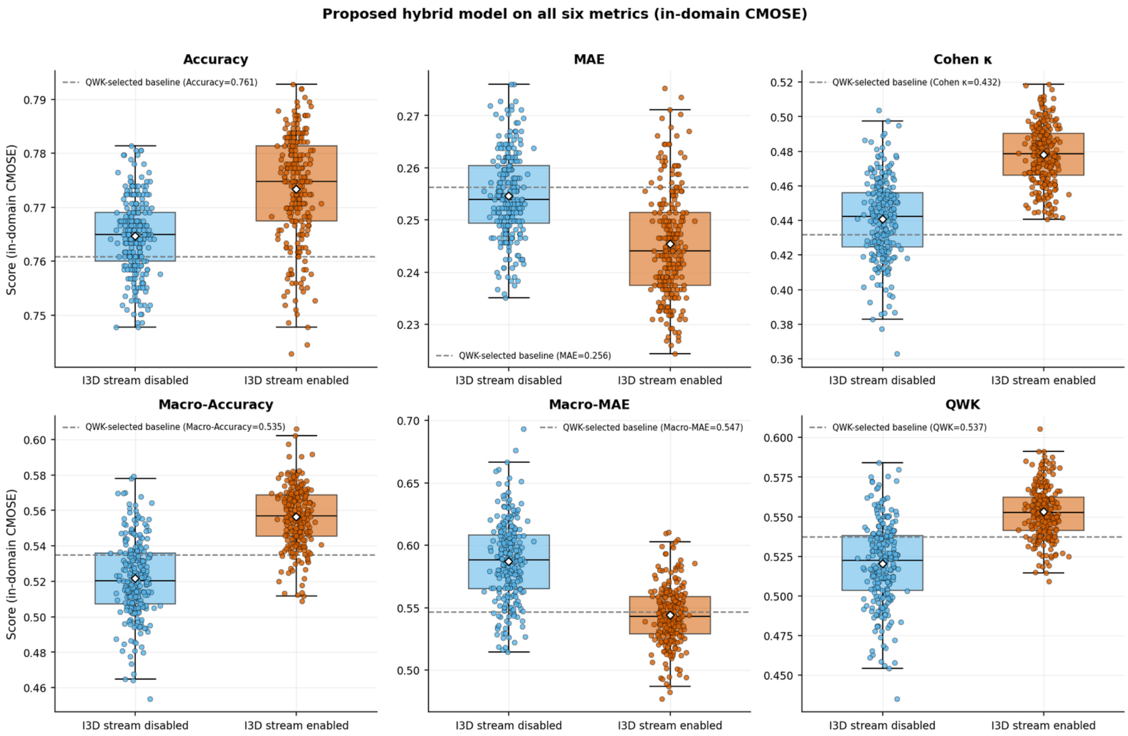



**Figure 4.1:** Ablation of the proposed hybrid model across all 243 per-group encoder configurations, with the I3D stream enabled and disabled, on all six metrics (in-domain CMOSE). The dashed line is the QWK-selected baseline (OpenFace TCN, cross-entropy).

weak cross-block agreement, and QWK again the only bridge, which reflects the metrics on this task, not any model family. Accuracy remains insensitive: every configuration falls within a narrow accuracy range around the baseline's 0.761 (top-left panel), so a model chosen on accuracy would see no architecture signal, while the QWK panel separates the two variants cleanly. The cross-domain collapse reproduces too: taken out of domain (Section 4.7.1), the proposed model's matrix has the same shape as the baseline matrix, a healthy diagonal (0.605 on CMOSE) and an off-diagonal collapse, with accuracy again masking it.

All three properties thus hold across a 32-fold larger model population, so the protocol carries over without amendment (RQ1's frame condition), and the experiments that follow can use it to judge the architecture.

## 4.4 Encoder Choice per Group

The design space varies five group→encoder choices at once; the aim here is to find which group's choice actually moves QWK (RQ2), and whether any preference is specific to one dataset or preserved under shift. Table 4.1 marginalises each group's encoder choice in both regimes, in-domain and over the unseen-target cells.

Only head pose shows a clear preference. In-domain it prefers a TCN (mean QWK 0.545, ahead of LSTM 0.541 and Transformer 0.523, a spread of 0.022), and

**Table 4.1:** Per-group marginal effect of encoder choice on mean QWK, in-domain (CMOSE) and pooled over the unseen-target cells.

| Group | Regime | TCN | Transformer | LSTM | Spread | Prefers |
|---|---|---|---|---|---|---|
| Gaze | In-domain | 0.539 | 0.537 | 0.534 | 0.006 | TCN |
| Gaze | Unseen targets | 0.093 | 0.095 | 0.094 | 0.002 | Transformer |
| Eye landmarks | In-domain | 0.538 | 0.537 | 0.535 | 0.003 | TCN |
| Eye landmarks | Unseen targets | 0.096 | 0.092 | 0.094 | 0.004 | TCN |
| Face landmarks | In-domain | 0.537 | 0.534 | 0.539 | 0.006 | LSTM |
| Face landmarks | Unseen targets | 0.092 | 0.096 | 0.093 | 0.004 | Transformer |
| Head pose | In-domain | 0.545 | 0.523 | 0.541 | 0.022 | TCN |
| Head pose | Unseen targets | 0.100 | 0.091 | 0.090 | 0.010 | TCN |
| Action units | In-domain | 0.536 | 0.537 | 0.537 | 0.001 | Transformer |
| Action units | Unseen targets | 0.093 | 0.096 | 0.093 | 0.003 | Transformer |

macro-MAE agrees, lowest under the TCN; the other four groups are insensitive to the encoder (QWK spread $\leq 0.006$). Under shift head pose again has the largest spread (0.010) and again prefers the TCN, while the other groups' spreads (0.002–0.004) flip winners arbitrarily, which is noise. The consistency across regimes shows the head-pose rule reflects the signal rather than a tuning accident: head dynamics such as nodding and turning away are temporally local and suit convolution (RQ2). The design space is otherwise flat, so most of the architecture's benefit must come from decomposing and fusing rather than from the encoder search, which the next experiment measures.

## 4.5 Effect of I3D Fusion

The baseline study predicted, from its error-overlap analysis (Appendix A, Section A.4.2), that fusing the I3D stream should help; this experiment tests the prediction. The design is paired: each I3D-stream-disabled configuration is matched with its exact I3D-stream-enabled twin (same five encoders), so the distribution of paired QWK differences is the I3D effect free of the configuration as a confounder, measured in three regimes (Table 4.2).

**Table 4.2:** Paired effect of the I3D stream on QWK, by evaluation regime.

| Regime | Pairs (n) | Mean $\Delta$QWK | Pairs improving (%) |
|---|---|---|---|
| Seen target | 972 | +0.060 | 94 |
| Cross-corpus (public) | 486 | -0.021 | 26 |
| Private (unseen) | 729 | +0.005 | 52 |

Where the test dataset is seen in training, fusion helps. Over the seen-target cells the mean paired gain is $+0.060$ QWK, and 94% of the 972 pairs improve.

In-domain on CMOSE the I3D-stream-enabled configurations are centred above the I3D-stream-disabled ones (median QWK 0.553 against 0.522, visible in Figure 4.1), and 82% of the 243 enabled configurations exceed the 0.537 baseline reference, against 26% of the disabled ones, so the gain belongs to the whole family (RQ3). Beyond the training datasets the gain disappears: cross-dataset the mean paired effect is $-0.021$ QWK with only 26% of pairs improving, and on the private set it is $+0.005$ with 52% improving and high variance. I3D's global appearance features carry dataset bias along, the same mechanism behind the I3D MLP's fall from second to fourth under shift (Appendix A, Section A.5.1). The rule (RQ3) is to enable the I3D stream when the target domain is represented in training and rely on the OpenFace groups otherwise; it is a rule about the family average, since the best single private-set model does have the I3D stream enabled but sits in one tail of a wide, zero-centred distribution.

## 4.6 Comparison with the Baseline

This experiment settles the in-domain comparison (RQ1): does the best configuration of the proposed model beat the best baseline (Appendix A), and where does any gain come from. Table 4.3 lists the top-5 configurations by QWK.

**Table 4.3:** Top-5 configurations of the proposed hybrid model, in-domain (CMOSE), sorted by QWK.

| Variant | Arch (gaze_eye_face_head_au) | QWK | Macro-Acc | Macro-MAE | Accuracy |
|---|---|---|---|---|---|
| I3D stream enabled | TCN_T_TCN_LSTM_T | 0.605 | 0.582 | 0.489 | 0.784 |
| I3D stream enabled | T_TCN_TCN_LSTM_T | 0.591 | 0.606 | 0.483 | 0.787 |
| I3D stream enabled | T_T_LSTM_LSTM_TCN | 0.591 | 0.563 | 0.505 | 0.781 |
| I3D stream enabled | T_TCN_T_TCN_T | 0.588 | 0.566 | 0.531 | 0.789 |
| I3D stream enabled | LSTM_LSTM_TCN_LSTM_T | 0.588 | 0.567 | 0.517 | 0.787 |

The best configuration improves on the baseline. It has the I3D stream enabled, `TCN_T_TCN_LSTM_T`, and every top configuration mixes at least two encoder types. On QWK, the selection metric, the best proposed-model configuration improves on the best baseline by $+0.068$ (0.605 versus 0.537), and the balanced secondaries corroborate it (macro-accuracy 0.582 versus 0.535, macro-MAE 0.489 versus 0.547), so the gain is real but modest.

The per-class recalls localise the gain. Relative to the best baseline, the proposed model improves recall on the ordinal extreme E3 (Highly-Engage) ($0.38 \rightarrow 0.51$) and on E1 (Disengage) ($0.45 \rightarrow 0.55$) while keeping E2 (Engage) near 0.90, and trades away a little of the rarest class E0 (Highly-Disengage) ($0.40 \rightarrow 0.37$), reallocating capacity toward the more frequent extreme. The in-domain improvement is therefore real but uneven, clearing the baseline on QWK and the balanced secondaries by strengthening the well-populated upper-engagement region and the middle disengaged class, while the ∼2–3%-frequency E0 class stays the main open challenge (RQ1). The next experiment shows the picture becomes more severe out of domain, and that the architecture's advantage over the baseline widens there.

## 4.7 Cross-Dataset Generalisation

The aim of the final experiment is the deployment question: how the proposed model transfers to unseen distributions, and whether the architecture gain and dataset combination reinforce each other there (RQ4). It proceeds from the 3×3 matrix to the private set, then asks whether the in-domain leaderboard predicts transfer.

### 4.7.1 Cross-Dataset Matrix

The proposed model's cross-dataset heatmap (Figure 4.2) shows the same pattern as the baseline matrix (Appendix A, Section A.5): a healthy diagonal and an off-diagonal collapse of QWK and the balanced secondaries alike, so the proposed model does not rescue cross-dataset transfer. Accuracy stays misleading, for the last time in the thesis's analysis: on DAiSEE→CMOSE the best QWK any of the 486 runs reaches is 0.10, yet those configurations reach accuracies up to 0.69. A second architecture family thus lands in the same place, so the metric frame is not an artefact of the single-encoder models.

Within the collapse, the proposed model is strongest in every cell. Figure 4.3 subtracts the baseline matrix (Figure A.5) from the proposed model's matrix cell by cell: the best proposed-model configuration outperforms the best baseline in all nine train×test cells, every cell of the ΔQWK matrix positive. The gains are smallest on the in-domain diagonal ($+0.068$ on CMOSE) and largest on the Private column ($+0.118$, $+0.050$, $+0.094$ from the three training sources). The improvement is a uniform shift of the whole matrix, not a single tuned point, the strongest form the family-level claim can take. Read column by column, the figure ties the two headline comparisons together: its CMOSE diagonal cell is the in-domain gain of Section 4.6, and its Private column is the private-set result developed next.

### 4.7.2 The Private Set

The main generalisation test is the private set: the 366 clips collected and hand-labelled for this thesis (Section 3.2.2), processed through the identical OpenFace/I3D pipeline and used for testing only. Because no model is trained on it, it is a test-only, in-the-wild set, with its own recording conditions and label prior (58% E2, 31% E3, 2.7% E0), and the only design lever is the training source. The Private column of Figure 4.3 and Table 4.4 report the best baseline and best proposed-model configuration per training source, where two effects combine.

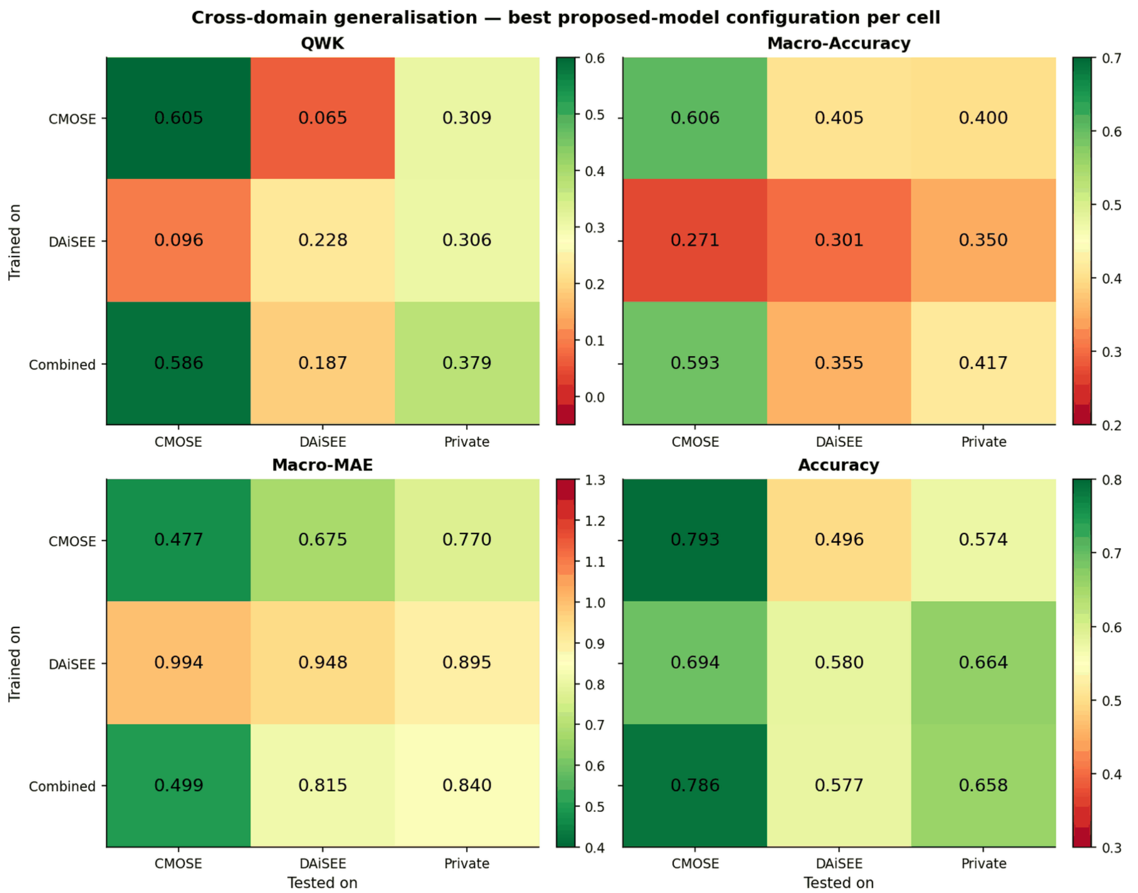



**Figure 4.2:** Cross-domain generalisation, best proposed-model configuration per cell (selected by QWK), on QWK, macro-accuracy, macro-MAE, and accuracy.

**Table 4.4:** Private set (test-only): best baseline vs best proposed hybrid model by training source.

| Train source | Best baseline (model/loss) | Baseline QWK | Baseline macro-MAE | Best proposed model (arch) | Proposed QWK | Proposed macro-acc | Proposed macro-MAE |
|---|---|---|---|---|---|---|---|
| CMOSE | openface_transformer/ce | 0.190 | 0.750 | I3D stream disabled, TCN_T_LSTM_TCN_T | 0.309 | 0.385 | 0.914 |
| DAiSEE | i3d_mlp/ce | 0.256 | 1.031 | I3D stream enabled, LSTM_LSTM_TCN_LSTM_LSTM | 0.306 | 0.350 | 0.966 |
| Combined | openface_transformer/ce | 0.285 | 0.997 | I3D stream enabled, T_T_T_LSTM_T | 0.379 | 0.385 | 0.877 |

(i) **Training source matters, and the combined dataset performs best.** For both model families, training on Combined generalises best to the private set (best baseline QWK: Combined 0.285 versus DAiSEE 0.256 versus CMOSE 0.190): combining mixed source datasets is the most effective simple route to an unseen target, as the baseline study showed (Appendix A).

(ii) **The proposed model generalises better, and helps more here than in-domain.** It outperforms the best baseline on QWK for every training source, and the gap is largest where it matters: under combined training, QWK 0.379 versus 0.285, a $+0.094$ improvement, wider than the $+0.068$ in-domain gain, so distribution shift amplifies rather than erodes the architecture's advantage. The best private-set model overall (selected on QWK alone) is the combined-trained proposed hybrid model with the I3D stream enabled, `T_T_T_LSTM_T`,

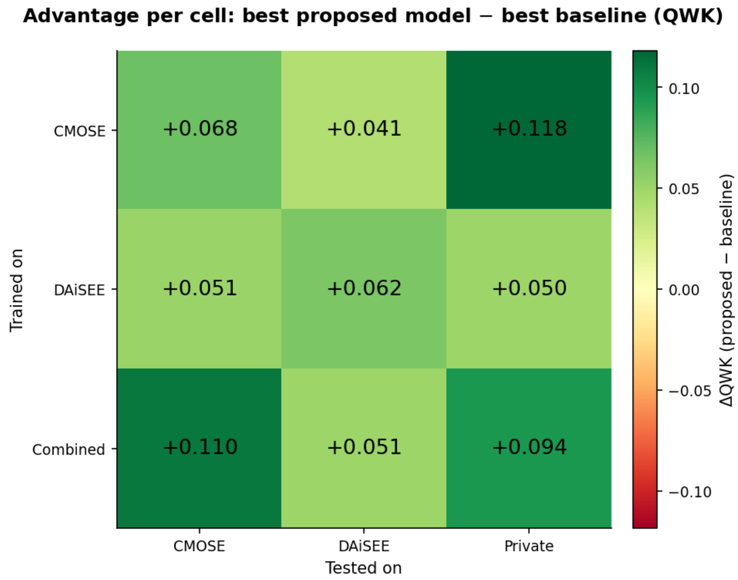



**Figure 4.3:** Cell-by-cell advantage of the proposed model: best proposed-model minus best baseline QWK over the 3×3 train×test matrix (Figure 4.2 minus Figure A.5).

at QWK 0.379, and the secondary macro-MAE agrees, the lowest of any model (0.877 against the best baseline's 0.997). One caveat in Table 4.4: the CMOSE-only baseline shows a deceptively low macro-MAE (0.75) despite the weakest QWK, by predicting conservatively around the majority classes, exactly the misleading reassurance that selecting on QWK alone is designed to avoid.

The confusion matrices localise where the gain comes from. Figure 4.4 compares the two combined-trained models on the private set: the proposed model captures the ordinal structure better, E2 recall rising from 0.46 to 0.72 with errors concentrated on the diagonal and its neighbours, where the baseline scatters E2 mass into E3 (0.44). Read as per-class F1, the same signature as in-domain (Section 4.6) recurs in sharper form: the proposed model lifts F1 on the well-populated E2 ($0.52 \to 0.70$) and E3 ($0.49 \to 0.56$) classes and, where the baseline is all but flat, gives the middle E1 class real mass ($0.09 \to 0.28$), while the rarest E0 class is traded all the way to zero ($0.07 \to 0.00$; all ten private E0 clips are misread). Out of domain the model concentrates its limited capacity on the engaged region even more sharply, and the rare disengaged extreme remains the thesis's open failure mode (RQ4).

### 4.7.3 In-Domain versus Transfer

The last question repeats, on the proposed model, the baseline test of whether in-domain ability predicts transfer (Appendix A, Section A.5.3). Each configuration's in-domain cell (trained and tested on CMOSE) is paired with its deployment cell (trained on Combined, tested on the private set), one point per

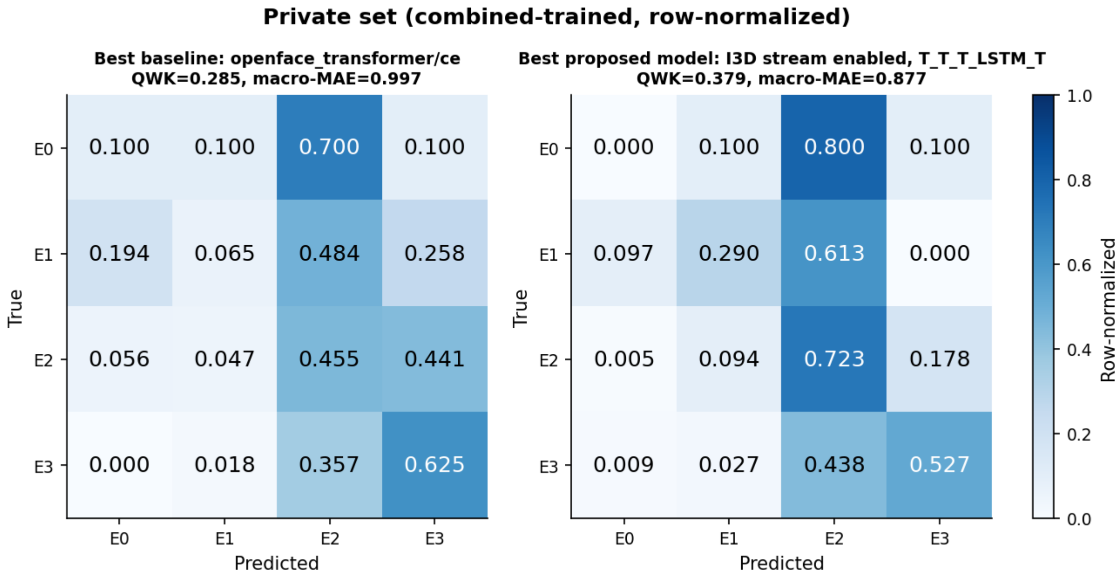



**Figure 4.4:** Private-set confusion of the combined-trained best baseline versus best proposed-model configuration (row-normalised).

configuration, and Figure 4.5 overlays the 486 proposed-model configurations on the baseline scatter.

The correlation stays weak ($\rho = 0.27$, significant only because $n = 486$), in the same range as the baseline relationship ($\rho = 0.31$): a configuration's in-domain CMOSE ability is, at most, a weak predictor of how it transfers. The architecture's contribution shows as position rather than slope: the proposed model's cloud lies above and to the right of the baseline configurations, reaching both higher in-domain QWK and higher private-set transfer, so the feature-group design lifts both ends together, while combined training remains the lever that places a model in that upper-right region. The private-set winner is therefore found by family-level robustness and combined training, not by reading the in-domain leaderboard.

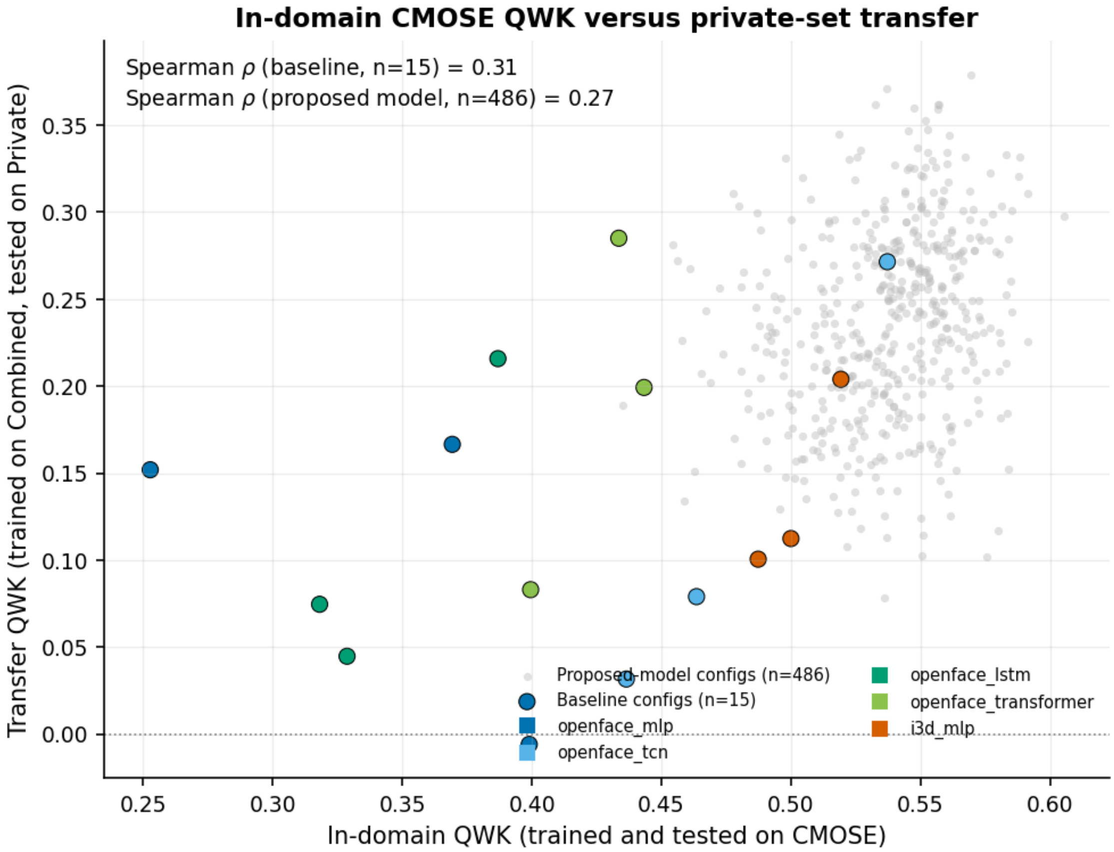



**Figure 4.5:** In-domain QWK (trained and tested on CMOSE) versus transfer QWK (trained on Combined, tested on the held-out private set); each configuration is one point pairing its own two runs. Spearman $\rho = 0.27$ for the 486 proposed-model configurations and $0.31$ for the 15 baselines.

## 4.8 Discussion

The design choices work for reasons visible in the results. Combining datasets improves generalisation by exposing the model to a more diverse set of distributions, reducing its tendency to overfit to either dataset. The proposed model's margin widens off-domain for the reason it was designed: each behavioural channel is encoded separately and forced to be discriminative through its auxiliary head, so the representation depends less on the distinctive features of any one dataset. The two feature families diverge under shift because the OpenFace descriptors live in subject-centred physical units that mean the same thing in any setup, whereas the I3D embedding encodes global appearance, what changes between datasets, so fusion adds signal only where the test statistics are represented in training.

The main result is a clear but moderate improvement. QWK 0.379 exceeds chance, every single-source model, and the best baseline $(+0.094)$, but in absolute terms it is moderate: with macro-MAE 0.877 the model captures the broad engaged-versus-disengaged trend yet is unreliable at fine distinctions, and the rarest class E0 fails in both regimes. The signal is usable as an aggregate, classroom-level measure but not to flag an individual disengaged learner without human confirmation. The open problem is one of data and objective, since no encoder assignment can recover a class the training distribution barely contains, which is why the remedies for imbalance in Chapter 5 are the natural next step.

## 4.9 Chapter Summary

The evaluation protocol held under the 486-configuration robustness check: the block structure of the metrics, the insensitivity of accuracy, and the cross-domain collapse all reproduced as task properties. Only head pose has a clear encoder preference (a TCN), so the benefit comes from decomposing and fusing rather than the encoder search (RQ2). I3D fusion is positive where the test dataset is seen in training, neutral-to-harmful beyond it (RQ3). In-domain the best proposed-model configuration outperforms the best baseline on QWK, the single selection metric (0.605 versus 0.537), with the secondary metrics agreeing (RQ1); out of domain it is strongest in every cell of the matrix, and reaches the best private-set model when trained on the combined dataset (QWK 0.379, $+0.094$ over the baseline, wider than in-domain), transferring better than the baselines at every in-domain level (RQ4), with E0 the main failure mode in both regimes. Chapter 5 draws the findings together, revisits the research questions, and outlines future work.

## 5.1 Summary

This thesis studied ordinal recognition of student engagement on a four-level scale, using the features OpenFace extracts from each frame and the features I3D extracts from the whole clip. It set out to improve recognition over standard single-encoder models and to describe the problem accurately, in particular its strong, uneven class imbalance and its behaviour under distribution shift.

The method is a complete, reproducible pipeline: feature extraction and leak-free preprocessing; the proposed hybrid model, which splits the 709-dimensional OpenFace features into five groups (gaze, eye landmarks, face landmarks, head pose, action units), encodes each group with its own temporal encoder, and optionally adds a global I3D motion stream while training every stream with a small auxiliary head; five single-encoder baselines; three loss functions; and a 3×3 cross-dataset evaluation protocol. The central idea was tested by training all $3^5 = 243$ per-group encoder choices, with the I3D stream enabled and disabled, and by a self-collected, test-only private set of 366 hand-labelled clips that serves as the deciding deployment test.

The evidence supports five conclusions.

(i) **The proposed hybrid model with the I3D stream enabled is the best in-domain model and, as a family, outperforms the strongest baseline.** The I3D-stream-enabled configurations are centred above the baseline QWK of 0.537 (median 0.553, 82% of the 243 clearing the baseline), and the best configuration reaches QWK 0.605 with a mixed TCN/Transformer/LSTM assignment (`TCN_T_TCN_LSTM_T`). The gain over the best baseline is real but small ($+0.068$ QWK), and the value lies in the framework and the full evidence, not in a jump in state of the art.

(ii) **Most groups are not affected by which encoder they get; only head pose shows a clear preference, for a TCN** (a 0.022 QWK spread across the three encoders, against $\leq 0.006$ for the other four groups). This simple design rule reflects the short, local nature of head motion.

(iii) **Engagement models do not generalise across datasets.** Off-diagonal QWK drops to near zero (CMOSE→DAiSEE 0.02) even when accuracy appears acceptable, so ordinal agreement metrics, not accuracy, must be reported on imbalanced engagement data.

(iv) **On the self-collected, test-only, in-the-wild private set, the three contributions combine.** The proposed model outperforms the best baseline in all nine train×test cells of the cross-dataset matrix. Because the private set is test-only, the only choice left is the training data; training on both datasets together with the proposed hybrid model gives the best result (QWK 0.379), outperforming the best baseline by $+0.094$ (against $+0.068$ in-domain) and surpassing every single-dataset-trained proposed model. This is the thesis's main practical result.

(v) **The rare E0 (Highly-Disengage) class is still the main open problem.** It is around 2–3% of the data, is traded away rather than solved by the proposed model in-domain, and drops to zero recall under distribution shift.

The architecture ablation rests on CMOSE because DAiSEE's in-domain ordinal signal is too weak (QWK 0.166) to tell architecture differences from label noise. CMOSE is therefore used for development and the private set for the deciding check.

## 5.2 Synthesis

Read together, the experimental results and the baseline study of Appendix A tell a single story whose lesson is not the one a benchmark paper usually reports. The three contributions of Section 1.4 (the architecture, the private dataset, and the cross-dataset study) are best read not as separate results but as one finding: each is modest on its own, and they reinforce one another where deployment is hardest.

The architecture is a small but clear in-domain improvement. The proposed hybrid model with the I3D stream enabled scores higher than the baseline, but the best configuration exceeds it by only $+0.068$ QWK, and most of the design space is not affected by which encoder it gets. Two things make it worth building. First, the fusion was measured before it was built: the clip-level agreement analysis of Appendix A showed that the best OpenFace model and the I3D baseline disagree more with each other than OpenFace models do among themselves, and that each family gets clips right that the other cannot, which the paired I3D ablation of Chapter 4 then confirmed in-domain. Second, four of five feature groups are not affected by their encoder and only head pose prefers a TCN, so the gain comes from splitting and fusing rather than a per-group encoder search, a more transferable design rule than a single tuned configuration. This is why the family-level effect, not any one cell, is reported as the result.

The private set turns the other two contributions from in-domain claims into a deployment test. The in-domain leaderboard predicts transfer only weakly ($\rho = 0.31$ for the baselines, 0.27 for the proposed model), so a model cannot be chosen for deployment from its in-domain score; a test-only, in-the-wild set recorded on real student webcams is the only place the combined claim can be settled. It is also where the architecture pays off most: the gap over the baseline grows from $+0.068$ in-domain to $+0.094$ on the private set, so the same decomposition that is a small in-domain gain becomes a representation that degrades more slowly under distribution shift, which, for a problem whose only real test is deployment, is what matters.

The cross-dataset study contributes negative results that are not side notes. The near-total collapse of cross-dataset transfer, the way accuracy hides it, and the steady failure on the rarest disengaged class are not minor limits of these particular models; they are aspects of the problem that the field leaves unmeasured by staying in-domain and reporting accuracy. Showing them with numbers, and showing that QWK, macro-accuracy, and macro-MAE make them visible, changes engagement recognition from beating a benchmark to reporting ordinal agreement and watching rare-class failure, which is the practice the rest of this chapter recommends.

## 5.3 Revisiting the Research Questions

The four research questions of Section 1.3 can now be answered directly.

**RQ1 (decomposition).** Decomposition helps. Encoding the five feature groups separately and fusing them outperforms encoding all 709 features with a single encoder: as a family the proposed model scores higher than the best baseline, and the effect holds across many configurations rather than one favourable configuration, with most of the I3D-stream-enabled configurations exceeding the baseline.

**RQ2 (encoder assignment).** Only head pose matters. Four of the five groups are not affected by whether their encoder is a TCN, a Transformer, or an LSTM (spread $\leq 0.006$ QWK); head pose alone shows a clear preference for a TCN. The design space is flat and a TCN is a safe default for every group, with head pose the one place the choice is worth making on purpose.

**RQ3 (motion fusion).** Yes in-domain, with a caveat under shift. When the model is tested on data like its training data, adding the clip-level I3D embedding helps: the paired comparison over the seen-target cells shows a mean gain of $+0.060$ QWK with $94\%$ of pairs improving, and the I3D-stream-enabled configurations give both the best in-domain model and the best private-set model. Under cross-dataset shift the same stream is neutral or slightly harmful on average (mean $-0.021$ on cross-public cells, $+0.005$ on the private set), because the I3D stream carries information useful but specific to each dataset. The fusion is recommended, but its gain should be claimed only for the setting where it holds.

**RQ4 (generalisation and losses).** Generalisation is poor and the metric choice is the key. Models do not transfer across datasets; off-diagonal ordinal agreement collapses while accuracy stays high and misleading. The in-domain score is at best a weak predictor of transfer ($\rho = 0.31$ for the baselines, 0.27 for the proposed model), so a model cannot be picked for deployment from its in-domain score alone. Among losses there is no overall winner (cross-entropy gives the best ordinal agreement while weighted and ordinal losses recover minority recall and balanced ordinal error), so the loss is a trade-off, but the cross-entropy-first order is the one in-domain preference that holds on every unseen target, which confirms, after the fact, the choice to establish it before development. QWK together with macro-accuracy and macro-MAE must be the main measure, and training on the combined datasets is the most useful tool for an unknown target.

## 5.4 Limitations

The findings should be read against several limits, each of which also motivates the future work below.

(i) **Single-annotator private labels.** The private set was labelled by a single annotator under the public-dataset rubric. Although the model suggestions shown during labelling were optional, the lack of a second annotator means no inter-annotator agreement can be reported, and the 366-clip size limits how reliable the private-set conclusions are, especially for the ten E0 clips, where a single wrong clip strongly affects the recall.

(ii) **Clip-level I3D features.** The I3D features are a single pooled 1024-dimensional vector per clip, so the I3D stream is an MLP over a global motion/appearance summary rather than a temporal encoder; a per-window I3D sequence might add more and would let an I3D temporal encoder model real dynamics.

(iii) **Small in-domain gain and a flat design space.** Most of the 243-configuration grid is not affected by which encoder it gets, and the best in-domain gain over the baseline is small ($+0.068$ QWK). The decomposition is a sound, interpretable framework rather than a large jump in performance.

(iv) **Weak DAiSEE signal.** The very low in-domain QWK on DAiSEE meant it could not be used for architecture development, so the cross-dataset conclusions rest mainly on CMOSE and the private set.

(v) **Fixed training recipe.** A single seed, learning rate, and encoder size were used throughout; without multi-seed repeats the reported gaps are point estimates rather than confidence intervals, and small differences between nearby configurations should not be over-interpreted.

## 5.5 Practical Recommendations

For practitioners building engagement-monitoring systems, the study gives three recommendations that do not depend on the specific architecture. First, select and optimise on ordinal agreement: make QWK the single metric that chooses models, report macro-accuracy and macro-MAE as secondary balanced checks, and treat accuracy as no more than a comparability number, because on these imbalanced ordinal labels accuracy can look healthy while the model has learned nothing useful about disengagement. Second, train on the combined datasets when the deployment data is unknown, which was the simplest tool that worked best for the private set and needs no change to the model. Third, anticipate and monitor rare-class failure: the most important class, E0, is also the rarest and hardest to detect, so a deployed system should treat low-confidence disengagement predictions with care and, where possible, leave the call to a human.

## 5.6 Suggestion for Future Works

(i) **Targeting the rarest class.** Combine the architecture with fixes aimed at the imbalance (ordinal losses tuned for E0, minority oversampling such as SMOTE [15], or cost-sensitive decision thresholds) to recover E0 recall without trading away E3. Because E0 is so rare, even small gains here would change the practical value of the system more than further QWK gains on the common classes.

(ii) **Domain adaptation.** The cross-dataset gap calls for explicit unsupervised domain adaptation or domain-invariant representation learning, rather than the plain dataset combination that already helps. A small amount of labelled target data could calibrate the model before deployment.

(iii) **Learned, not listed, group encoders.** Replace the discrete 243-configuration search with a differentiable architecture search or a gating mechanism that picks per-group temporal operators end-to-end, so the model finds which encoder suits which group on its own.

(iv) **Stronger and richer motion features.** Re-extract per-window I3D (or a modern video backbone) so the motion stream carries timing, and add audio/speech features, which CMOSE also provides, as extra feature groups within the same split-then-fuse framework.

(v) **Frequency-domain analysis of engagement (exploratory).** A natural extra direction, left strictly as future work, is to study the temporal frequency of facial signals, under the idea that engaged behaviour is low-frequency and steady while disengaged behaviour is higher-frequency and restless, for example through short-time-Fourier-transform features or convolutions over the frequency axis.

(vi) **Stronger evaluation.** Multi-annotator labelling of a larger private set, with reported inter-annotator agreement, and multi-seed training with confidence intervals would tighten every gap reported here and turn point estimates into statistically grounded claims.

## 5.7 Closing Remarks

Automatic engagement recognition is appealing exactly because online learning makes the webcam the main link between student and teacher, but the same setting makes the problem difficult: the labels are ordinal, severely imbalanced, subjective, and specific to each dataset. This thesis showed that keeping the structure of the facial features, by splitting them into feature groups and encoding each on its own terms, gives a sound, interpretable model that, combined with a motion stream and trained on combined datasets, generalises best to unseen, in-the-wild data. It also showed that the field's usual reliance on accuracy hides a near-total failure to transfer across datasets, which ordinal agreement metrics reveal. Detecting the rarest, most important disengaged behaviour under distribution shift remains open, and is the most important next step.

## REFERENCE

[1] C.-H. Wu **andothers**, "CMOSE: Comprehensive Multi-Modality Online Student Engagement Dataset with High-Quality Labels," **in***Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) Workshops* 2024, **pages** 4636–4645. DOI: 10.1109/CVPRW63382.2024.00466

[2] A. Gupta, A. D'Cunha, K. Awasthi **and** V. Balasubramanian. "DAiSEE: Towards User Engagement Recognition in the Wild," **urlseen** 26 **june** 2026. **url**: https://arxiv.org/abs/1609.01885

[3] T. Baltrušaitis, A. Zadeh, Y. C. Lim **and** L.-P. Morency, "OpenFace 2.0: Facial Behavior Analysis Toolkit," **in***13th IEEE International Conference on Automatic Face and Gesture Recognition (FG)* 2018, **pages** 59–66. DOI: 10.1109/FG.2018.00019

[4] O. Çopur, M. Nakip, S. Scardapane **and** J. Slowack, "Engagement Detection with Multi-Task Training in E-Learning Environments," **in***Image Analysis and Processing – ICIAP 2022* **jourser** Lecture Notes in Computer Science, **volume** 13233, Springer, 2022, **pages** 411–422. DOI: 10.1007/978-3-031-06433-3_35

[5] S. Malekshahi, J. M. Kheyridoost **and** O. Fatemi. "A General Model for Detecting Learner Engagement: Implementation and Evaluation," **urlseen** 26 **june** 2026. **url**: https://arxiv.org/abs/2405.04251

[6] F. M. Shiri, E. Ahmadi, M. Rezaee **and** T. Perumal, "Detection of Student Engagement in E-Learning Environments Using EfficientNetV2-L Together with RNN-Based Models," *Journal on Artificial Intelligence*, **jourvol** 6, **number** 1, **pages** 85–103, 2024. DOI: 10.32604/jai.2024.048911

[7] L. Zhang, X. Zheng, X. Chen **and** L. Cui, "Three-Dimensional View Relationship-Based Context-Aware Emotion Recognition," *IEEE Transactions on Neural Networks and Learning Systems*, **jourvol** 36, **number** 7, **pages** 13 567–13 578, 2025. DOI: 10.1109/TNNLS.2024.3476249

[8] I. Qarbal, N. Sael **and** S. Ouahabi, "Student's Engagement Detection Based on Computer Vision: A Systematic Literature Review," *IEEE Access*, **jourvol** 13, **pages** 140 519–140 545, 2025. DOI: 10.1109/ACCESS.2025.3596885

[9] M. M. Santoni, T. Basaruddin **and** K. Junus, "Convolutional Neural Network Model based Students' Engagement Detection in Imbalanced DAiSEE Dataset," *International Journal of Advanced Computer Science*

*and Applications (IJACSA)*, **jourvol** 14, **number** 3, **pages** 617–626, 2023. DOI: 10.14569/IJACSA.2023.0140371

[10] M. Tan **and** Q. V. Le, "EfficientNetV2: Smaller Models and Faster Training," **in***Proceedings of the 38th International Conference on Machine Learning (ICML)* **jourser** Proceedings of Machine Learning Research, **volume** 139, 2021, **pages** 10 096–10 106.

[11] J. Carreira **and** A. Zisserman, "Quo Vadis, Action Recognition? A New Model and the Kinetics Dataset," **in***Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)* 2017, **pages** 4724–4733. DOI: 10.1109/CVPR.2017.502

[12] S. Bai, J. Z. Kolter **and** V. Koltun. "An Empirical Evaluation of Generic Convolutional and Recurrent Networks for Sequence Modeling," **urlseen** 26 **june** 2026. **url**: https://arxiv.org/abs/1803.01271

[13] S. Hochreiter **and** J. Schmidhuber, "Long Short-Term Memory," *Neural Computation*, **jourvol** 9, **number** 8, **pages** 1735–1780, 1997. DOI: 10.1162/neco.1997.9.8.1735

[14] A. Vaswani **andothers**, "Attention Is All You Need," **in***Advances in Neural Information Processing Systems (NeurIPS)* 2017, **pages** 5998–6008.

[15] N. V. Chawla, K. W. Bowyer, L. O. Hall **and** W. P. Kegelmeyer, "SMOTE: Synthetic Minority Over-sampling Technique," *Journal of Artificial Intelligence Research*, **jourvol** 16, **pages** 321–357, 2002. DOI: 10.1613/jair.953

[16] L. Hou, C.-P. Yu **and** D. Samaras, "Squared Earth Mover's Distance-based Loss for Training Deep Neural Networks," *arXiv preprint arXiv:1611.05916*, 2016.

[17] D. P. Kingma **and** J. Ba, "Adam: A Method for Stochastic Optimization," **in***International Conference on Learning Representations (ICLR)* 2015.

[18] J. Cohen, "A Coefficient of Agreement for Nominal Scales," *Educational and Psychological Measurement*, **jourvol** 20, **number** 1, **pages** 37–46, 1960. DOI: 10.1177/001316446002000104

[19] C. Spearman, "The Proof and Measurement of Association between Two Things," *The American Journal of Psychology*, **jourvol** 15, **number** 1, **pages** 72–101, 1904. DOI: 10.2307/1412159

[20] M. G. Kendall, "A New Measure of Rank Correlation," *Biometrika*, **jourvol** 30, **number** 1–2, **pages** 81–93, 1938. DOI: 10.1093/biomet/30.1-2.81

# APPENDIX A. BASELINE MODELS

## A.1 Experimental Setup

Every result in this appendix comes from one apparatus. The five single-encoder baselines of Section 3.3 are each trained under the three losses of Section 4.2.1 on each of the three training sources, optimised identically (Adam, fixed seed, early stopping on validation loss; Section 4.2.2) and evaluated on the held-out test split under the $3\times3$ cross-dataset protocol of Section 4.2.3. The in-domain experiments (Sections A.2–A.4) use the fifteen CMOSE-trained, CMOSE-tested configurations (five architectures $\times$ three losses); the cross-dataset experiment (Section A.5) uses the full matrix.

Each run is scored on all six metrics of Section 2.3.4, but the reporting is deliberately reduced to follow the protocol. The first experiment (Section A.2.1) settles on QWK as the single metric that selects every best model; from then on the two balanced secondary metrics, macro-accuracy and macro-MAE, accompany each headline result to describe how a model treats the rare classes, while accuracy, micro-MAE, and Cohen's kappa are reported only where they are themselves the object of study, accuracy where its unreliability is the point (Section A.5.2) and Cohen's kappa where prediction agreement is (Section A.4). Two figures deliberately show the complete six-metric grid for reference (Figure A.3 here and Figure 4.1 in Chapter 4); every other result follows this reduced reporting.

Two aggregation conventions hold across this appendix and the experimental results (Chapter 4). A winner is the single configuration with the highest QWK over the tunable axis (here the loss), ties broken stable-first: because QWK is the only selection metric, a reported best model never depends on any other number. A claim about a family of configurations is instead reported as a mean and a distribution over the design grid. Every configuration is trained once with a fixed seed (Section 4.2.2), so a reported mean measures robustness to design choices, not noise across seeds.

## A.2 Evaluation Protocol

Before the baselines can be ranked, three parameters of the protocol must be fixed: the metric that selects models, the dataset the models are developed on, and the loss they are trained with. Each is settled by one experiment below. Two of the arguments need out-of-domain evidence and are stated here as claims, then proved in Section A.5.2.

### A.2.1 Selecting the Primary Metric

The first parameter to fix is the metric that selects models, because on an ordinal, imbalanced task that choice is not neutral. Engagement levels lie on a line, and 69.4% of CMOSE is the single class E2 (Engage) (Table 3.1); accuracy ignores both properties, rewarding majority-class collapse and counting a one-level miss the same as a three-level miss, whereas QWK is chance-corrected and weights every error by the squared distance between predicted and true levels (Section 2.3.4). To test the consequence, the fifteen in-domain baseline configurations are scored on all six metrics, and two questions are put to them: does the leaderboard winner depend on the metric, and do the metrics rank the configurations the same way.

The winner depends on the metric. Table A.1 reports which of the fifteen configurations each metric declares best in-domain, and no two camps agree: QWK picks the OpenFace TCN under cross-entropy (0.537), both macro metrics pick the I3D MLP under the ordinal loss (macro-accuracy 0.635, macro-MAE 0.458), and accuracy, micro-MAE, and Cohen's kappa all pick the I3D MLP under cross-entropy (0.773, 0.244, 0.436).

**Table A.1:** Winning baseline configuration by metric (in-domain CMOSE).

| Metric | Best model | Loss | Value |
|---|---|---|---|
| QWK | openface_tcn | CE | 0.537 |
| Macro-Accuracy | i3d_mlp | Ordinal | 0.635 |
| Macro-MAE | i3d_mlp | Ordinal | 0.458 |
| Accuracy | i3d_mlp | CE | 0.773 |
| MAE | i3d_mlp | CE | 0.244 |
| Cohen $\kappa$ | i3d_mlp | CE | 0.436 |

The metrics also fall into two blocks bridged only by QWK. Figure A.1 reports the Kendall rank correlation between every pair of metrics over the fifteen configurations: a micro block (accuracy and micro-MAE, $\tau = 0.98$, dominated by the majority class) and a macro block (macro-accuracy and macro-MAE, $\tau = 0.89$) that are almost unrelated to each other (cross-block $\tau \approx 0.01$–$0.10$). QWK is the only metric with moderate-to-strong agreement with both blocks ($\tau = 0.49$–$0.60$).

The choice of metric therefore changes the result and must be argued, not defaulted. This thesis adopts QWK as its single primary metric, the compromise sensitive to ordinal distance without being dominated by the majority class: the winner is the argmax of QWK, ties broken stable-first, with no second metric consulted. The remaining five are secondary. The decisive argument against

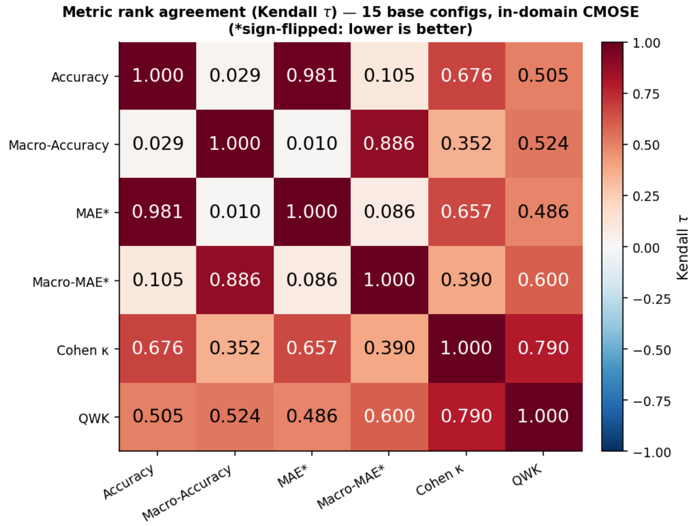



**Figure A.1:** Rank agreement (Kendall $\tau$) between the six metrics over the fifteen baseline configurations, in-domain on CMOSE (MAE metrics sign-flipped so positive $\tau$ means agreement).

accuracy, that it cannot even detect the collapse of a model's ordinal signal under shift, needs out-of-domain evidence and is deferred to Section A.5.2. With the metric fixed, the next question is which dataset to develop on.

### A.2.2 Selecting the Development Dataset

A development dataset is usable only if models rise meaningfully above chance on it, since a dataset where they do not cannot separate good architectures from bad. The two in-domain cells are compared for this property in Table A.2, each with its best model.

**Table A.2:** Best in-domain result per dataset (CMOSE vs DAiSEE).

| In-domain dataset | Best model | Loss | QWK | Macro-Acc | Macro-MAE | Accuracy |
|---|---|---|---|---|---|---|
| CMOSE | openface_tcn | CE | 0.537 | 0.535 | 0.547 | 0.761 |
| DAiSEE | i3d_mlp | CE | 0.166 | 0.289 | 1.226 | 0.548 |

Only CMOSE clears the bar. The best in-domain model reaches QWK 0.537 on CMOSE but only 0.166 on DAiSEE (macro-accuracy 0.535 versus 0.289, macro-MAE 0.547 versus 1.226), while DAiSEE's raw accuracy of 0.548 hides the failure, a first sign of how misleading accuracy can be. CMOSE is therefore the development dataset; DAiSEE re-enters as a transfer source and target in Section A.5 and in Chapter 4, where its difficulty is informative rather than obstructive. With the metric and the dataset fixed, the last parameter is the loss.

### A.2.3 Selecting the Development Loss

The loss is the axis that trades the primary metric against the balanced secondary ones, so it must be fixed before the architecture is judged. For every baseline, the loss is swept from cross-entropy through weighted cross-entropy to the ordinal loss, in-domain on CMOSE, and QWK is plotted against the balanced metrics (Figure A.2).

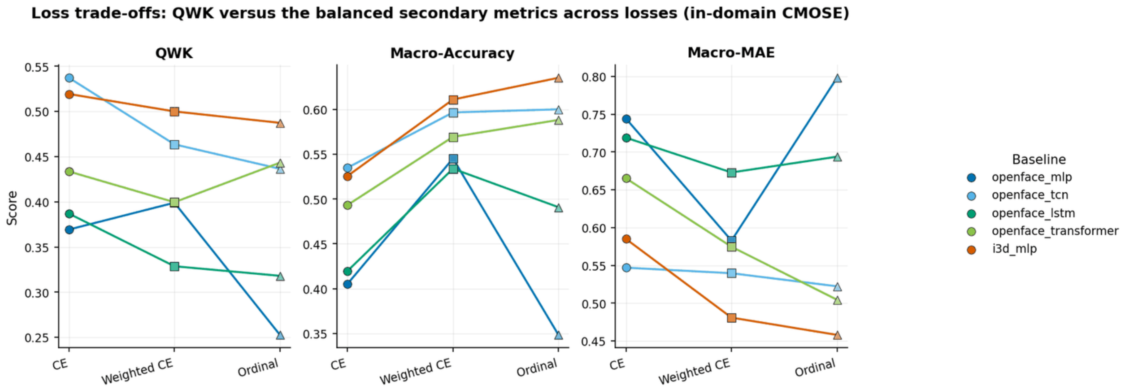



**Figure A.2:** QWK against the balanced secondary metrics (macro-accuracy and macro-MAE) for every baseline across the three losses, in-domain on CMOSE.

The trade is Pareto: rebalancing the loss raises macro-accuracy and improves macro-MAE while QWK falls for four of the five baselines, so no loss is best on QWK and on the balanced metrics at once. For the I3D MLP, switching from cross-entropy to the ordinal loss lifts macro-accuracy from 0.526 to 0.635 and improves macro-MAE from 0.585 to 0.458 (the best balanced figures of any baseline) while QWK falls from 0.519 to 0.487. Cross-entropy is the development loss because it maximises QWK, the selection metric, and isolates the effect of the architecture from that of the loss (best-architecture QWK by loss: CE 0.537, weighted CE 0.500, ordinal 0.487); the proposed hybrid model of Chapter 4 is consequently trained with it. That this CE $>$ weighted CE $>$ ordinal ranking survives domain shift is the second deferred claim, closed in Section A.5.2. With the three parameters fixed, the baselines can be ranked.

## A.3 In-Domain Results

The aim of this experiment is to set the reference level the proposed architecture must exceed. The five single-encoder baselines are trained and tested in-domain on CMOSE under each of the three losses (four on the OpenFace descriptor: an MLP, a TCN, an LSTM, and a Transformer; and one on the I3D clip vector, the I3D MLP); Figure A.3 reports the complete grid on all six metrics for reference.

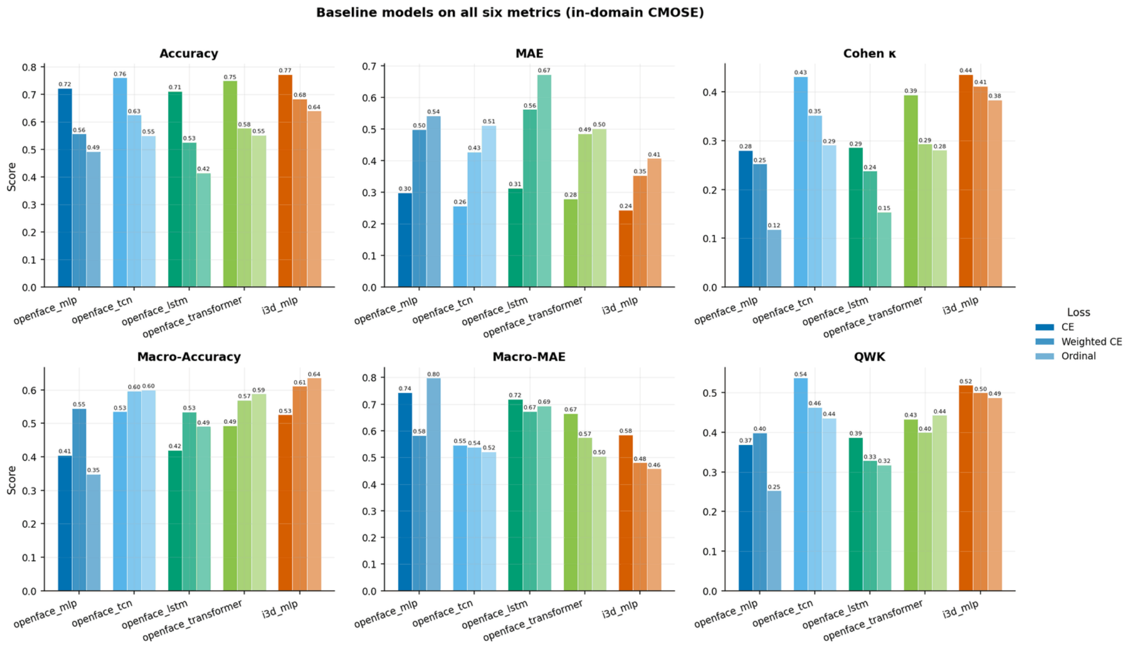



**Figure A.3:** The five baseline models under the three losses on all six metrics, in-domain on CMOSE.

On QWK, the selection metric, the strongest baseline is the OpenFace TCN under cross-entropy at QWK 0.537: this is the reference level. The grid adds two structural observations. The loss axis determines on which metric family a model is strongest, cross-entropy on the micro and agreement metrics, the rebalanced losses on the macro metrics that weight the rare disengaged classes equally. The architecture axis determines which kind of feature is strongest: the OpenFace TCN's per-frame facial dynamics (a head turning away, gaze drifting off-screen) separate the rarer disengaged levels and rank highest on QWK, while the I3D MLP, over a single pooled appearance vector, aligns with the majority E2 class and ranks highest on every micro metric. On QWK the two score almost the same (0.537 versus 0.519). Two models this close, yet built on disjoint feature families, pose a question the score table cannot answer: are they interchangeable, or do they succeed on different clips?

## A.4 Prediction Agreement

Scores summarise how often a model is right, not which clips it is right on. This experiment descends to individual predictions on the 1,221 in-domain CMOSE test clips to ask whether models with similar scores make the same decisions, first across the whole grid, then for the two feature families. Cohen's kappa, the agreement statistic, is the metric of study here.

### A.4.1 Agreement between Configurations

The fifteen configurations agree far less than their clustered scores suggest. Figure A.4 reports the pairwise Cohen's kappa between every pair of predictions: mean pairwise agreement is only $\kappa = 0.27$–$0.39$, and sharing the architecture ($\kappa = 0.391$) or the loss ($\kappa = 0.374$) raises it only mildly over sharing neither ($\kappa = 0.273$). The configurations are different hypotheses about the data, not reparametrisations of one another.

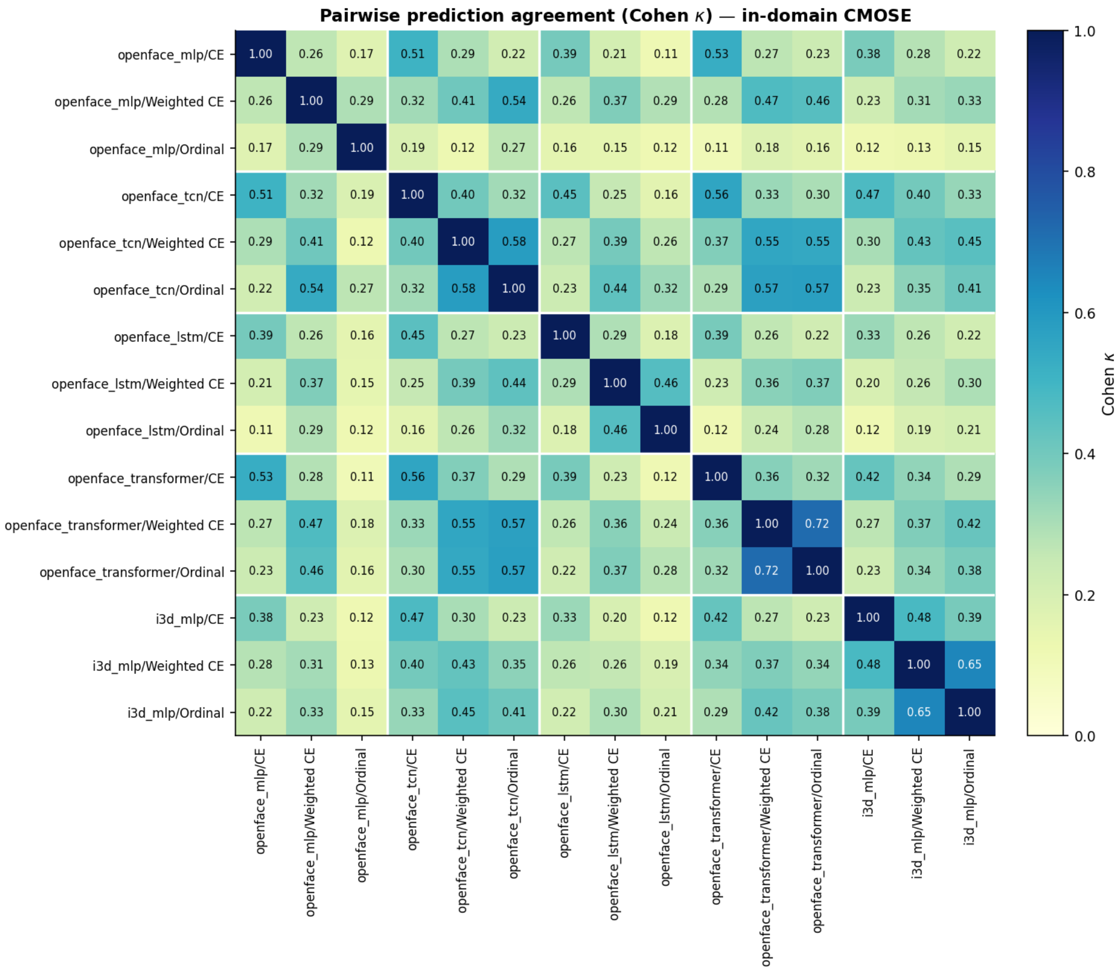



**Figure A.4:** Pairwise prediction agreement (Cohen's $\kappa$) between all fifteen baseline configurations on the in-domain CMOSE test set, in model-major blocks.

The consequence is headroom. The best single configuration classifies 77.3% of the test clips correctly, but at least one of the fifteen is correct on 97.6% (the oracle), so only 2.4% of clips defeat every model. Plain majority voting over the fifteen lands at 73.5%, below the best single model, because the errors are too decorrelated for unweighted voting to use. Claiming the headroom therefore requires a learned combination of complementary views of the input, which is the design premise of the proposed hybrid model of Chapter 4: it fuses separately encoded behavioural streams instead of averaging finished predictions.

### A.4.2 OpenFace versus I3D

The two feature families scored almost the same, yet differ in nature: OpenFace is a per-frame, structured account of facial behaviour (gaze, eye and face landmarks, head pose, action units), while I3D is a single clip-level vector of global appearance and motion. Whether their close scores mean redundancy or complementarity is the question. Table A.3 compares their best cross-entropy models: the OpenFace TCN is strongest on QWK and on both balanced secondary metrics, so by scores alone the I3D MLP appears merely redundant.

**Table A.3:** Best OpenFace encoder versus the I3D MLP (in-domain CMOSE, cross-entropy).

| Feature family | Model | QWK | Macro-Acc | Macro-MAE |
|---|---|---|---|---|
| OpenFace (behaviour descriptors) | openface_tcn | 0.537 | 0.535 | 0.547 |
| I3D (appearance/motion) | i3d_mlp | 0.519 | 0.526 | 0.585 |

The agreement map says otherwise. In Figure A.4 the I3D MLP's row is the palest in the grid: its agreement with the OpenFace TCN is among the lowest of any pair, despite the two having the closest scores. The clip-level decomposition confirms it: against the OpenFace TCN, the I3D MLP has the largest exclusive-correct share of any pair in the grid (8.6% of test clips only I3D gets right) and the smallest both-wrong share (15.3%), and the pair's oracle accuracy is 84.7%, $+7.4$ points over the best single model and the largest pair gain in the grid. The behaviour descriptors and the appearance features look at different evidence and fail on different clips. This is the empirical motivation for fusing the two streams in the proposed hybrid model of Chapter 4, a motivation the score table alone could never supply.

## A.5 Cross-Dataset Generalisation

In-domain numbers are not the deployment question. This experiment trains on one dataset and tests on another, with three aims: measure whether any ordinal signal survives a change of dataset, close the two arguments the protocol deferred, and ask whether in-domain strength predicts generalisation.

### A.5.1 The 3×3 Matrix

The 3×3 heatmaps of Figure A.5 are the central generalisation result for the baselines, and they show a cross-dataset collapse. On the diagonal QWK is healthy (CMOSE 0.54), macro-accuracy is well above chance (CMOSE 0.64), and ordinal error is low (CMOSE macro-MAE 0.46). Off the diagonal all three collapse together: CMOSE→DAiSEE QWK $= 0.02$ and DAiSEE→CMOSE $= 0.05$ (chance-level agreement), macro-accuracy falls toward the 0.25 chance floor, and macro-MAE

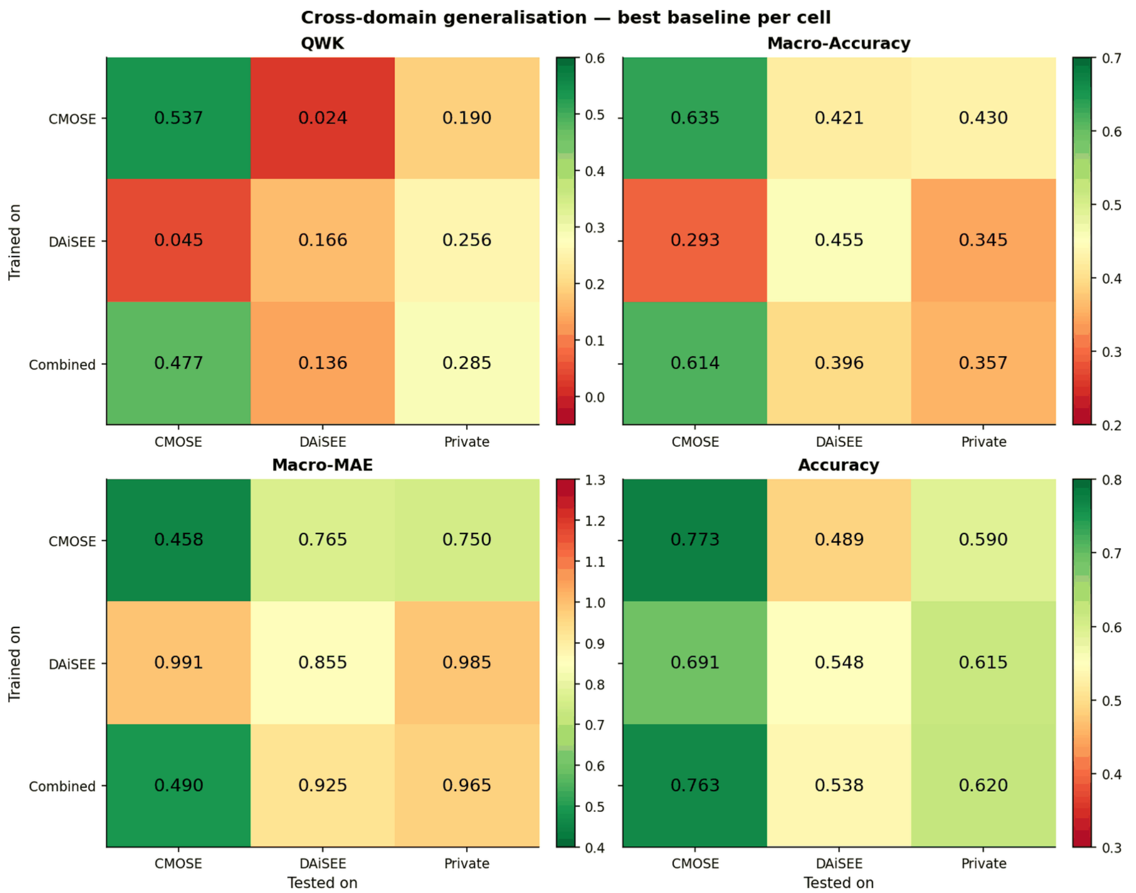



**Figure A.5:** Cross-domain generalisation, best baseline per cell (selected by QWK), on QWK, macro-accuracy, macro-MAE, and accuracy.

inflates toward a full class. The collapse is roughly symmetric, which rules out one dataset being a superset of the other and points to a real shift in both label priors and recording conditions.

The architecture ranking also changes out of domain. Comparing, per architecture, the in-domain QWK with the mean QWK over the unseen-target cells, the in-domain order TCN 0.537 $>$ I3D 0.519 $>$ Transformer 0.443 $>$ MLP 0.399 $>$ LSTM 0.387 becomes TCN $\approx$ Transformer $>$ LSTM $>$ I3D $>$ MLP on unseen targets (Figure A.6). The I3D MLP falls from second to fourth: its global appearance features bind to the dataset, while the OpenFace behaviour descriptors carry more signal that holds across datasets.

Combining the source datasets is the most effective simple fix for an unknown target. On the unseen Private column, training on the Combined dataset gives the best private-set QWK (0.285), exceeding DAiSEE-only (0.256) and CMOSE-only (0.190), while keeping near-in-domain CMOSE performance (0.477). The combined dataset covers a wider range of label priors and recording conditions than either

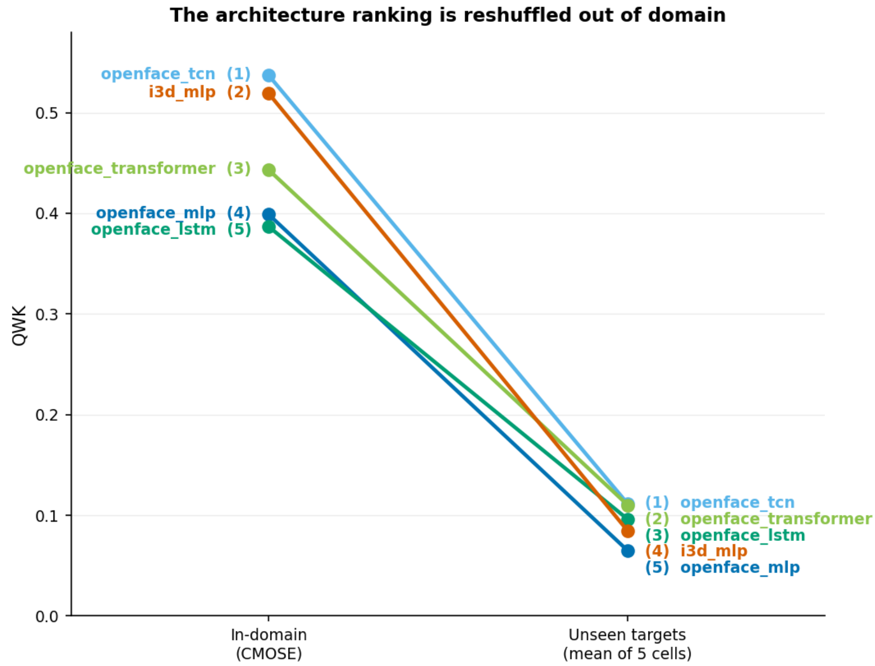



**Figure A.6:** Architecture QWK ranking, in-domain (left; trained and tested on CMOSE) versus the mean over the five unseen-target cells (right), best over loss. One line per architecture, with the rank (1 = best) at each end.

alone, so a model trained on it transfers better to an unseen target. Chapter 4 combines this with the architecture contribution on the decisive private set (RQ4).

### A.5.2 Closing the Loop: Metric and Loss Reliability

The cross-dataset matrix supplies the out-of-domain evidence the protocol deferred. Accuracy does not detect the collapse, which completes Section A.2.1: in the accuracy panel of Figure A.5, CMOSE→DAiSEE shows a deceptively reasonable 0.49 and DAiSEE→CMOSE 0.69, in the same cells where the QWK panel reads 0.02 and 0.05, because predicting near the majority class scores well on any imbalanced target. A metric that looks normal while the ordinal signal is gone is misleading, so accuracy is the least reliable metric here and QWK, the only bridge between both blocks, the most.

The loss ranking is preserved under the shift, which completes Section A.2.3. On the unseen-target cells the best-architecture QWK by loss follows the same CE $>$ weighted CE $>$ ordinal order as in-domain. The loss is the only factor in the grid whose ranking transfers (the architecture ranking reshuffles), so choosing cross-entropy in-domain remains right out of domain, confirming after the fact the choice to establish it as the development loss.

### A.5.3 In-Domain versus Transfer

The third aim is to ask whether the in-domain leaderboard predicts transfer. Pairing, for each of the fifteen configurations, its in-domain QWK (trained and tested on CMOSE) with its deployment transfer QWK (trained on Combined, tested on the private set) gives one point per configuration; the Spearman rank correlation is only a non-significant $\rho = 0.31$. A strong in-domain score is therefore at best a weak signal of private-set transfer, so the in-domain leaderboard is not a reliable basis for selecting a model for an unknown target, and selection must instead rely on combining datasets. Chapter 4 re-examines the same relationship on the 486-configuration population of the proposed hybrid model, where it stays weak ($\rho = 0.27$, Figure 4.5).

## A.6 Chapter Summary

The evaluation protocol established QWK as the single primary metric that selects the best model (with macro-accuracy and macro-MAE the reported secondaries), CMOSE as the development dataset, and cross-entropy as the development loss. Under it, the strongest baseline is the OpenFace TCN at QWK 0.537, the reference level the proposed hybrid model must exceed. The prediction-level experiment showed the configurations agree far less than their scores suggest (oracle 97.6%) and that the OpenFace and I3D families fail on different clips, the direct motivation for fusing them. The cross-dataset experiment showed baselines do not transfer to a new dataset, confirmed the metric and loss choices, and made dataset combination the most effective simple fix. Chapter 4 takes up whether learning to fuse the complementary streams outperforms these baselines, and whether the architecture gain and dataset combination reinforce each other on unseen data.